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# Mapping the landscape of Artificial intelligence for serious games in Health: An enhanced meta review

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ABSTRACT

This paper presents an enhanced meta-review of artificial intelligence (AI) applications in serious games (SGs) for health, focusing on adaptation, personalisation, and real-time data processing to improve rehabilitation outcomes. A systematic review methodology, following PRISMA-ScR guidelines, was employed to analyse studies from 2017 to 2025, identifying the AI algorithms most frequently used to personalise gaming experiences, improve patient engagement, and increase treatment efficacy. Our proposal categorises the algorithms based on their role in the game environment, which can either be game control or user assessment. Game control algorithms adapt the game environment and difficulty, while user assessment algorithms gather information about the player's state, such as performance, mood, or physiological data, to evaluate the treatment progress. The review also examines the growing trend of using multimodal data as input for machine learning models.

Results show that a few well-known algorithms, such as Decision Trees (DT), Artificial Neural Networks (ANN), Fuzzy Logic (FL), Naïve Bayes (NB), and Support Vector Machines (SVM), are frequently used. However, there is a clear distinction in their purposes: while FL is typically applied to game control tasks, SVMs are mainly used for user assessment.

This review offers valuable insights for researchers in the field, providing a comprehensive overview of the suitability of different AI algorithms for various tasks in SGs, with a particular focus on personalisation and motivation.

List of Abbreviations

| Abbreviation | Full Term | Abbreviation | Full Term |
|---|---|---|---|
| AdaBoost | Adaptive Boosting | IBI | Interbeat Intervals |
| ADAM | Adaptive Moment Estimation | INN | Integer Bounded Neural Network |
| AI | Artificial Intelligence | JSON | JavaScript Object Notation |
| ANN | Artificial Neural Network | k-NN | K-Nearest Neighbours |
| ANFIS | Adaptive Neuro-Fuzzy Inference System | k-means | K-means Clustering |
| AR | Augmented Reality | LDA | Linear Discriminant Analysis |
| BFGS | Broyden-Fletcher-Goldfarb-Shannon | LLM | Large-scale Language Modelling |
| Bi-LSTMs | Bidirectional Long Short-Term Memory | LR | Logistic Regression |
| BVP | Blood Volume Pulse | LSTM | Long Short-Term Memory |

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| Abbreviation | Full Term | Abbreviation | Full Term |
|---|---|---|---|
| CBR | Case-Based Reasoning | MCI | Mild Cognitive Impairment |
| CatBoost | Categorical Boosting | MTOM | Machine Theory of Mind |
| CNN | Convolutional Neural Network | ML | Machine Learning |
| CN2 | CN2 Algorithm (Rule Induction Algorithm) | MLP | Multilayer Perceptron |
| DDA | Dynamic Difficulty Adjustment | NB | Naïve Bayes |
| DL | Deep Learning | NPC | Non-Playable Character |
| DNN | Deep Neural Network | PCG | Procedural Content Generation |
| DRL | Deep Reinforcement Learning | PD | Parkinson's Disease |
| DT | Decision Tree | HOG | Histogram of Oriented Gradients |

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(*continued*)

| Abbreviation | Full Term | Abbreviation | Full Term |
|---|---|---|---|
| ε-greedy | Epsilon-Greedy Strategy | HOG | Histogram of Oriented Gradients |
| ECG | Electrocardiography | PEM | Player Experience Modelling |
| EDA | Electrodermal activity | PSO | Particle Swarm Optimisation |
| EEG | Electroencephalography | RBF | Radial Basis Function |
| EMG | Electromyography | RF | Random Forest |
| EM | Expectation-Maximization | RFE | Recursive Feature Elimination |
| Extra Tree | Extremely Randomised Trees | RL | Reinforcement Learning |
| FFO | Firefly Optimisation | RNN | Recurrent Neural Network |
| FL | Fuzzy Logic | RQ | Research Question |
| fNIRS | Functional Near-Infrared Spectroscopy | RT | Regression Tree |
| FSM | Finite State Machine | sEMG | Surface Electromyography |
| GA | Genetic Algorithm | SGs | Serious GameSurface |
| GAN | Generative Adversarial Networks | SNG | Serious Neurogame |
| GNB | Gaussian Naïve Bayes | SVM | Support Vector Machine |
| GOAP | Goal-Oriented Action Planning | TPOT | Tree-based Pipeline Optimisation Tool |
| HBM | Hierarchical Bayesian Modelling | VR | RealityTree-based |
| HMM | Hidden Markov Model | WoK | Web of Knowledge |
| HRV | Heart Rate Variability | XAI | Explainable Artificial Intelligence |

## 1. Introduction

Video games appeared first in the 1960s and experienced a boom in the late 1980s and early 1990s (Donovan). Nowadays, the video game market has gained a stable position in our lives. Currently, the video game industry has three major platforms to branch out from: mobile gaming, PC gaming, and console gaming. Due to their popularity, video games became interesting for serious applications, i.e. to be used in multiple sectors like health care, professional training, learning and education, military and defence, often in combination with smart devices and sensors. This genre of video game, not mainly developed for entertainment purposes, is called Serious Games (SGs) (Abt, 1987).

The research and employment of SGs grew especially in the medical sector offering many fields of application like screening ((Vallejo et al., 2015), (Tong et al., 2016a), (Tong et al., 2016b), (Zygouris et al., 2020)), diagnosis ((Brown et al., 2014), (Tsionas et al., 2020), (Mezrar & Bendella, 2022)), prevention ((Wiemeyer & Kliem, 2012), (Fleming et al., 2015)), or rehabilitation ((Tannous et al., 2015), (Martins et al., 2020)). Nevertheless, there remains a long way to fully integrate SGs into treatments, because due to a lack of long-term studies and experience in real environments, no standardisation took place ((Damaševičius et al., 2023), (Domingo et al., 2023)). Meanwhile, research is advancing quickly: the current boom of applying Artificial Intelligence (AI) to many areas is also entering SGs, as it could offer new ways and possibilities, e.g. the personalisation of therapies, which has been proved to enhance the therapeutic effect ((Damaševičius et al., 2023), (Domingo et al., 2023), (Jung et al., 2020), (D'Aprile et al., 2019)).

AI was formally introduced in the 1950s. In the 1990s, with the emergence of Artificial Neural Networks (ANN) and Deep Learning (DL) technologies of the 21st century, the global industry came to a deeper understanding of the immense importance of AI technologies for a new wave of industrial transformation. In the medical field, the complexity and rapid growth of healthcare data suggest that AI will be more and more widely used in this field, covering a wide range of applications such as personalised screening, diagnosis, prognosis, monitoring, risk modelling, drug discovery, or treatment response prediction.

Serious games (SGs) that combine gaming technology with specialised applications are expected to drive significant change in the medical field in the future. With the continued development of deep learning technologies such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), SGs will be able to acquire complex biomedical data (e.g., physiological signals and medical images) from users in real time and transmit it to back-end intelligent systems for analysis ((Petsani et al., 2022), (Patiño et al., 2025), (Ascari et al., 2020, pp. 100–110)). This data can then be used by AI models to identify key features and patterns to aid in diagnosis, training and rehabilitation interventions. For example, CNNs can identify and annotate pathological features in images in medical training scenarios, providing users with an immersive and efficient teaching experience and reducing human error ((Mienye et al., 2025), (Chen et al., 2018, pp. 397–402)). RNNs are suited to processing continuous physiological data, such as heart rate, electromyography (EMG) (Gonzalez-Vazquez et al., 2024), or electroencephalography (EEG), to provide real-time biofeedback (Kim, Yang, et al., 2021) for rehabilitation protocols to support dynamic adjustments and personalised optimisation. This paradigm shift is expected to not only improve the accuracy and interactivity of medical education but also fundamentally reshape the traditional treatment and rehabilitation process.

By analysing the user's health status, rehabilitation needs, and personal preferences through AI, it is possible to create a customised SG with tailored environments and rehabilitation programs that not only enhance user engagement through appealing and challenging settings but also improve commitment and overall effectiveness of the rehabilitation process. Through well-selected game mechanics, it is possible to motivate patients to adhere to their program, as monotony or stress could be avoided or reduced ((Goršič et al., 2017), (Baert et al., 2011), (DeSmet, 2014)).

Furthermore, systems based on generative artificial intelligence with large-scale language modelling (LLM) could open up new possibilities for adaptive rehabilitation in the future. These technologies could dynamically modify game environments and interaction modes in real time, based on users' physiological and psychological feedback. Virtual agents could simulate real-world healthcare professionals or rehabilitation coaches to deliver personalised guidance and encouragement, enhancing immersion and user engagement ((DeCock et al., 2020), (Brom et al., 2008), (Emmerich & Masuch, 2016)). Through affective computing and natural language processing, AI could also sense users' emotional states ((Grodniewicz & Hohol, 2024)) and adapt its interaction style to reduce anxiety and resistance during therapy (Mundargi et al., 2025), making rehabilitation more human-centred and effective (Liszio et al., 2017).

Also, combined with real-time data analysis by AI, an SG could continuously monitor participants' physiological parameters, exercise performance, and rehabilitation progress. This real-time feedback would allow the medical professionals to more accurately assess the patient's status and make timely adjustments to better meet the patient's needs. This not only could improve the efficiency of treatment, but also reduce the risks that may arise during the treatment ((Robert, 2014), (Antoniou et al., 2020), (Hickman et al., 2015), (Sosa et al., 2015)).

Through AI's deep learning analysis of this data, SGs can adaptively adjust the game difficulty and feedback mechanism based on the user's real-time status. For example, when it detects that the heart rate is too high or the EEG shows a high state of stress, the system can automatically reduce the difficulty of the task or adjust the training tempo to avoid overfatigue or stress overload (Gonzalez-Vazquez et al., 2024), (Darzi et al., 2021), (Nasri, 2025a). Conversely, if the system recognises that rehabilitation is progressing well, it can appropriately increase the challenge to ensure the effectiveness and motivation of the training. As wearable devices and sensor technology continue to advance, this real-time data-driven adaptive capability will become even more precise and efficient, thus improving the quality and safety of rehabilitation training.

Nevertheless, auditing the systems is essential to ensure patient safety. Both the patient and the therapist have to trust the system, being able to interpret its output and understand why a model makes certain decisions. To assure that, methods such as LIME (Local Interpretable Model-agnostic Explanations) (Ribeiro et al., 2016) and SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017) were created to explain the predictions of any classifier, and these methods will grow together with the advancements of AI usage in general. In (Clement et al., 2023), the authors illustrated how higher complexity of algorithms leads to harder explainability. They presented a thorough literature analysis to derive valuable insights into the development process of XAI (Explainable AI) software by organising it along the five typical phases inherent to a software development process.

The motivation for our research stems from the absence of in-depth review articles analysing the AI techniques applied to SGs. Although there are many review articles about the use of SGs in health, their focus is mainly on certain purposes (e.g. rehabilitation, care, policy or medical education) and generally the types of games, the users or the effectiveness on certain diseases is analysed, but not the technical background. Therefore, we identified only two comprehensive reviews that highlight how AI techniques are applied, the first published at the beginning of AI-usage for SGs (2017) and the second one five years later (2022). In 2017, Frutos-Pascal et al. (Frutos-Pascual & García Zapirain, 2017) exhaustively examined AI algorithms guided by three key research areas in game-AI, introduced as "flagships" in 2012 (Yannakakis, 2012). The flagships are an interesting focus to classify the high amount of works published but the analysis is made in a narrative style, which makes it difficult to get an overview of which techniques are more frequently used and for which cases. The authors also analysed the "AI-Usage", i.e. applications of the different AI techniques, classifying them into decision-making and machine learning algorithms, but without providing classification tables. In 2022, Abd-alrazaq et al. (Abd-alrazaq et al., 2022) reviewed AI-driven SGs in healthcare, but algorithms were not analysed in detail. Different game characteristics were summarised in narrative form, and although more information was given in a table, many of the cited articles included multiple AI methods and were not classified, so the reader has to refer to the original papers to find out specific data. Despite significant advances in research in the intervening years in this field, we found no other reviews focusing on the techniques applied, so given the rapid development of diverse AI methods for various applications, we believe it is time to clarify the landscape. We aim to offer a comprehensive overview of the different combinations of algorithms and tasks, helping researchers identify the most appropriate algorithm for their needs. We focus particularly on the customisation and adaptation of games using AI to enhance their efficiency and effectiveness, especially in the context of rehabilitation.

To this end, introduce a simplified classification of AI algorithms in SGs into two functional categories: **user assessment** and **game control**. This framework aligns conceptually with the flagships model proposed by Yannakakis (Yannakakis, 2012), which identifies four key research areas in game AI: Player Experience Modelling (PEM), Large-Scale Game Data Mining, Procedural Content Generation (PCG), and new perspectives in NPC AI. Our proposed categories consolidate these research areas according to their operational roles in SGs.

- **User assessment** encompasses the analysis of both objective and subjective data generated by or about the player. This includes in-game metrics (e.g. scores, interaction patterns), physiological and behavioural signals (e.g. heart rate, movement), and self-reported information (e.g., emotions, motivation, or survey responses). This category maps closely to **PEM**, which has been defined as "the study and use of AI techniques for the construction of computational models of the experience of players" (Frutos-Pascual & García Zapirain, 2017). PEM integrates perspectives from user modelling, affective computing, experimental psychology, and human-computer interaction. These models often combine multimodal data to infer player states or experiences. ***Large-Scale Game Data Mining***, which involves analysing extensive datasets to assess player behaviour and gameplay quality, also falls within this category. In health-related SGs, this includes AI methods used to monitor rehabilitation progress or assess clinical outcomes.
- **Game control** refers to algorithms designed to adapt or generate game content in response to player needs or system objectives. This includes **PCG**, defined as "the study and development of algorithms that generate content automatically" (Yannakakis, 2012). Such content spans game elements like levels, maps, narratives, quests, and challenges, all of which shape the player experience. Additionally, emerging approaches to **NPC behaviour**—aimed at enhancing realism, believability, and adaptive interaction—are also subsumed under this category.

Fig. 1 tries to visualise the relationship between the four flagship areas and our two categories: to achieve an optimal working game, it has to be re-adjusted continuously by a regulatory cycle that consists of sensing (user assessment) and acting (game control). Changes in the game influence the experience and motivation as a consequence. The measurement of physiological data, performance, affective state, etc., gives information on the success of those changes, and readjustments should be made to keep the player in a flow state and maintain or augment their motivation. Here, the part of the user assessment plays a double role, apart from the information necessary for readjustments, it delivers valuable data about the player's health status and achievements in therapy (the arrow pointing out of the circle stands for the information flow to the therapist).

To address the gap in the existing literature and clearly distinguish the two primary purposes for employing AI in SGs, we formulated one general and two more specific research questions (RQs).

**RQ1:** Which AI algorithms have been explored for use in SGs and gamification in the health sector, and how has their usage evolved?
**RQ2:** Which AI algorithms are predominantly used for tasks related to **game control**, to personalise the player experience, meet the needs of different users and improve the immersion and therefore the therapeutic effectiveness of games?
**RQ3:** Which AI algorithms focus on data evaluation for **user assessment** (particularly involving multimodal data), to: a) predict disease, health status, the success of rehabilitation, etc., and b) obtain input data for game control, i.e., feedback for continuous adaptation and readjustment of the game content to maintain and enhance motivation?

Our analysis is performed in two steps: As a starting point, we first present a meta-review that follows our classification. Subsequently, new research not included in the first part is reviewed. The results of both are finally joined to present an in-depth review that covers publications from nearly 25 years.

The rest of the paper is organised as follows. In section 2, the review methodology is detailed, section 3 presents the results of the meta-review and the follow-up review in two sub-sections, and section 4 compares and discusses the results regarding our RQs. Finally, section 5 concludes the article by summarising the most important findings.

## 2. Materials and methods

This review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) (Tricco et al., 2018). The meta-review analyses the most relevant reviews about SGs, games in general, gamification etc.,

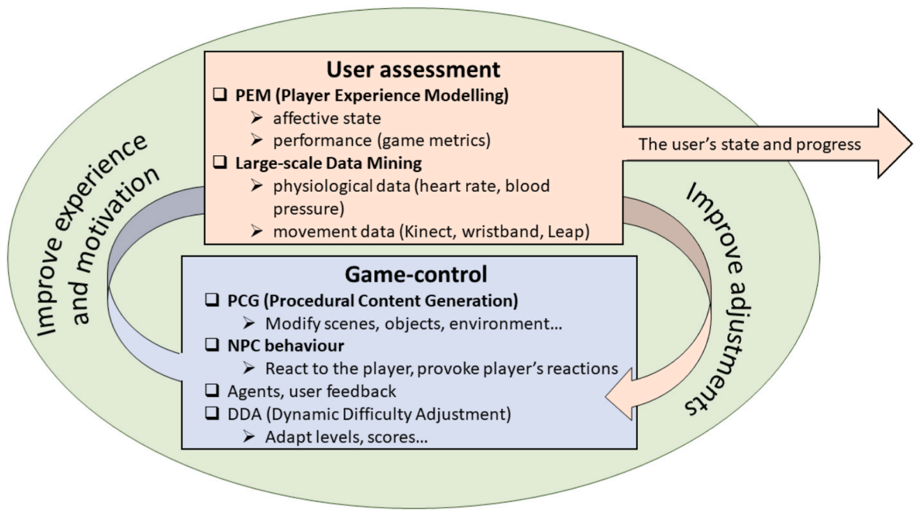


**Fig. 1.** Connection between the flagships identified by Yannakakis et al. (Yannakakis, 2012) and the two areas of AI usage: user assessment and game control.

focusing on different aspects as the use of AI, but also the use of other algorithms that aim to adapt or personalise the games. In that way, we tried to enhance the view around the two only reviews found about AI in SGs mentioned earlier ((Frutos-Pascual & García Zapirain, 2017), (Abd-alrazaq et al., 2022)) and built a basis for the second part, which is a scoping review focused on the most recent research in this area. Nevertheless, we allowed an overlap of some years between the newest review and the starting year to include important works that possibly were not included in the reviews.

### 2.1. Determination of search criteria and selection process

The search strategy was led by the research questions and was performed according to the following steps.

a) Database selection:

For this study, we used the most popular and prestigious online databases like Web of Knowledge (WoK), IEEE Xplore, Scopus, and PubMed. Review articles were only searched in WoK, as our criterion here was to find the most reliable and representative studies, i.e. those published in high-ranked journals and therefore indexed in WoK.

b) Specification of a relevant time frame:

2017 to 2023 for review articles and 2019 to 2023 for research articles.

c) Definition of Keywords (with variations): `game, serious game, exergame, gamification, gaming, artificial intelligence, AI, machine learning, ML, deep learning, DL, adaptation, personalization, individualization, health, prediction, screening, diagnosis, multimodal`
d) Establishment of search queries and combinations:
e) Firstly, some basic short queries were defined covering every aspect of the research questions:
   - A. (game* OR gami*)
   - B. (AI OR "artificial intelligence" OR ML OR "machine learning" OR DL or "deep learning")
   - C. (adapt* OR personal* OR individuali*)
   - D. ("serious game*" OR exergame*)
   - E. (health* OR rehabilitation)
   - F. (predict* OR screen OR diagnosis)
   - G. multimodal

Then, different combinations of these queries were composed according to our two targets:

The search for **review papers** was restricted to the Titles (TI) and Document type = Review Article, with the following combination of queries (see visualisation in Fig. 2).

Q1. A = reviews about games or gamification
Q2. A AND B = sub-set of Q1 using AI, ML or DL
Q3. A AND C = sub-set of Q1 dealing with personalisation or adaptivity
Q4. A AND D = sub-set of Q1 for serious games and exergames
Q5. A AND D AND E = sub-set of Q4 for health

The search for **research papers** was performed in all fields, and the following combination of queries (see visualisation in Fig. 3).

Q6. D AND E = articles about serious games or exergames in health or for rehabilitation
Q7. D AND E AND B = sub-set of Q6 using AI, ML or DL
Q8. D AND E AND B AND C = sub-set of Q7 dealing with personalisation or adaptivity
Q9. D AND E AND B AND F = sub-set of Q7 for disease prediction
Q10. D AND E AND G = sub-set of Q6 using multimodal data

f) Removal of duplicates:

Articles found multiple times for different queries or databases were identified and removed.

g) Revision of titles and abstracts:

Publications that were not relevant to healthcare, AI, and serious gaming were excluded. Reviews were only considered for analysis if they dealt with technical proposals and concrete AI algorithms used for games and gamification purposes. Research articles were excluded if treating studies, teaching AI programming, game theory or if named "game changer" or similar.

h) Refinement after full read:

The remaining reviews and research articles were passed through a full read by two authors. Some reviews have been excluded if the method was not rigorous enough or if they did not provide relevant data to answer our review questions. Research articles have been excluded if the contribution was not clear or if the algorithms used were not

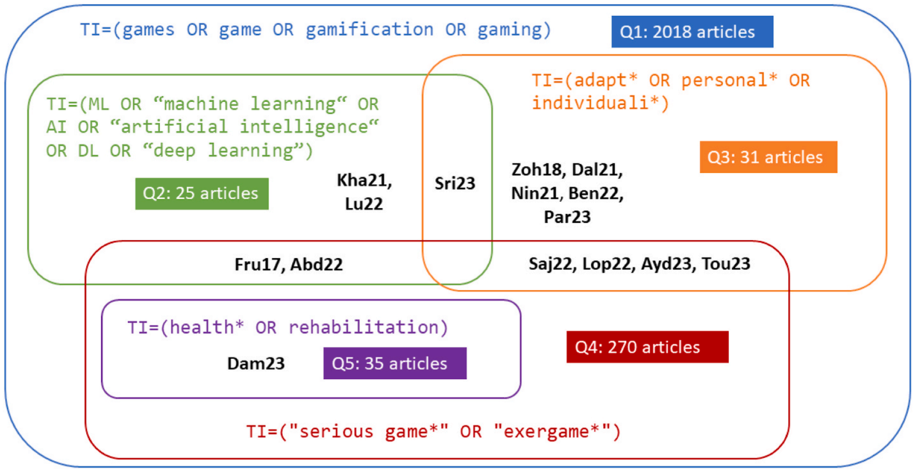


**Fig. 2.** Illustration of the search process and results obtained for the meta-review. Mnemonics represent the selected articles.

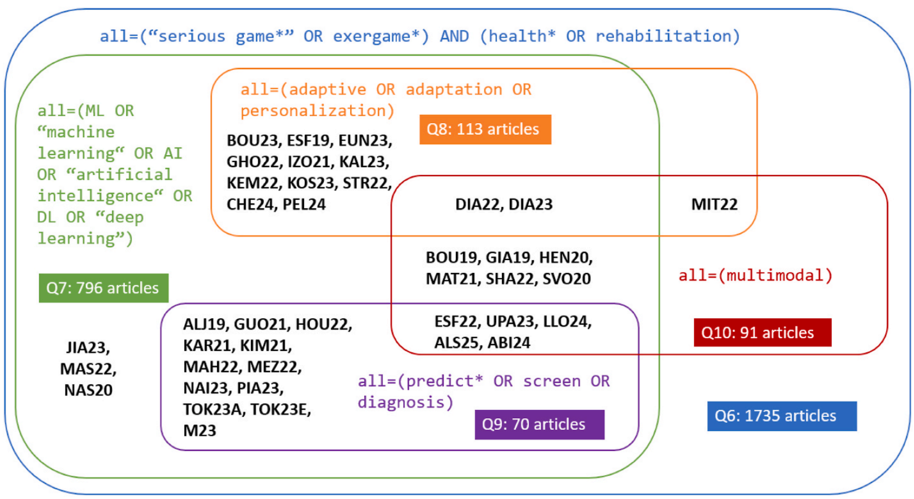


**Fig. 3.** Illustration of the search process and results obtained for the scoping review. Mnemonics represent the selected articles.

mentioned or insufficiently explained.

The whole search and analysis process has been organised via an Excel data sheet, which we provide as complementary information for the reader.

### 2.2. Data extraction and analysis

For the analysis of review articles (meta-review), we established the following order:

First, we studied Dam23 (Damaševičius et al., 2023), the newest comprehensive meta-review about SGs and Gamification in Healthcare, to get a general overview and detect if part of our RQs had already been answered.

We then analysed the remaining 14 reviews in order of relevance, starting with the five reviews that focused on AI, as this was our primary concern and was not the main focus of Dam23 (Damaševičius et al., 2023) (section 3.2.1). Within those articles, our review order began with SGs, followed by general games, and finally, adaptiveness and personalisation in SGs, general games or gamified applications. Next, we reviewed the nine articles treating adaptation and personalisation without a special focus on AI (section 3.2.2), starting with adaptive SGs and concluding with adaptive general games. In each of these steps, we always started with the oldest publication.

Method of analysis for classification of algorithms:

In each article, the objective and research questions were inspected first, then the results were analysed by classifying the algorithms mentioned according to their purpose, i.e. if they were created for tasks related to game control or user assessment. We counted an algorithm as game control when AI was used to change the game content or difficulty, e.g. game flow, levels, scenarios, NPC behaviour, or to make decisions that influence the gaming experience. The user's performance or something similar could be used as input parameters, if it were data produced inside the game, i.e., known by the game. On the contrary, algorithms were classified as user assessment in the case that the user's physiological and affective state, behaviour, interaction, experience etc., was measured or predicted, i.e. the adaptation of the game was based on data that was not available inside the game. The type of game adaptation was mostly fixed in these cases, e.g., the result of the user assessment was used to decide between predefined difficulty levels.

The analysis of research articles not included in the reviews or published after those included in the meta-review was performed according to the focus on personalisation of the rehabilitation experiences, analysis of the players' performance and disease prediction. To follow up with the outcome of the meta-review, results were grouped according to user assessment and game control-oriented algorithms and their frequency was added in the last two columns of the results table.

For better mnemonic referencing, we opted to name each article according to a code composed of the first three letters of the surname of the first author, followed by the year of publication. To distinguish the articles included in both sections, in the meta-review part, only the first letter is uppercase, while in the scoping review part, all three letters are uppercase.

## 3. Results

### 3.1. Search results

Fig. 2 illustrates the outcome of the different combinations of the queries Q1 to Q5. From a total of 2018 review articles found about games or gamification (Q1, blue cloud), 25 treated AI (Q2, green cloud), 31 were about personalisation or adaptivity (Q3, orange cloud) and 270 were about serious games (Q4 red cloud). From the latter, 35 could be filtered out for the health sector (Q5, violet cloud). The figure also shows the mnemonics of the articles we selected from each query and visualises which were repeatedly found (overlaps). All 15 finally selected articles are listed with ID, title and reference in Table 1, in the order of analysis.

Fig. 3 visualises the results of the search process for Q6 to Q10, to find the most recent research articles and enhance the landscape drawn by the meta-review. Q6 (articles about SGs and exergames in health or for rehabilitation) was part of all searches and was not executed separately, so the number of 1735 articles found has been estimated.

For Q7 (using AI), we obtained 796 search results in the initial search. 93 articles were identified as duplicates or similar publications of the same study and removed. We read through the titles and abstracts of the remaining 703 articles, removing publications that were not relevant to healthcare, AI, and serious gaming, and obtained 210 papers that required a full-text read-through. After evaluating and summarising the findings and conclusions of those publications, we excluded studies that only mentioned AI or serious gaming in the text to ensure that all articles were relevant to the topic. Finally, 40 articles were selected for this study. Q8 filtered the former results to find works that aim to adapt or personalise the games; 14 articles fulfilled this requirement. With Q9, the results of Q7 were filtered for disease prediction, and 17 articles were found here.

With the last query, Q10, we wanted to find articles dealing with multimodal data, but not necessarily using AI, as to widen the range for initial works and new ideas, because this is still a growing field. Therefore, this search was executed independently and yielded 91 initial results. After removing duplicates and similar publications of the same study, we read through the titles and abstracts of the remaining articles and found ten of the articles we selected in the former search process, and only one new article worth including that does not use AI.

Table 2 lists all selected articles with their mnemonic codes, references, titles, and the category of the proposed or applied algorithms.

In the following two sections, the results of the review articles and the newest research articles are presented and analysed separately. During analysis, all algorithms mentioned were added to a results table (Table 3) according to the most common ML categories generally found in the literature (supervised learning (SL), unsupervised learning (UL), and reinforcement learning (RL)), and the category "others" (O) to classify algorithms that did not fit into the former ones. Furthermore, each category was split into classes, subdivided into "specific implementations" if those were mentioned. If the authors of a work did not identify the types or genres of the algorithms applied in a cited article, in certain cases, we scanned them to obtain this information. However, since this was a very time-consuming task, it was only carried out in the most interesting cases. Columns "G" and "U" identify the number of algorithms found to be used for game control and user assessment.

### 3.2. Analysis of review articles (meta-review)

The following sections are ordered according to the methodology described in section 2.2 and following the order of Table 1.

After giving a general view of SGs and exergames in health according to Dam23 (Damaševičius et al., 2023), section 3.2.1 considers the usage of AI, and section 3.2.2 analyses the articles found about adaptiveness and personalisation in games and SGs without focus on AI.

The review presented by Dam23 (Damaševičius et al., 2023) comprises a profound and recent survey of 53 works in the domain of SGs and Gamification published between 2017 and 2021. Their focus was on extracting the current research trends in SGs and Gamification, visualising the various study methodologies employed and examining the

**Table 1**
List of the review articles analysed in the meta-review.

| Topic | | ID | Title | N | Years |
|---|---|---|---|---|---|
| **Overview of SG for health** | | Dam23 (Damaševičius et al., 2023) | Serious Games and Gamification in Healthcare: A Meta-Review | 53 | 2017–2021 |
| **Focus on AI** | **SG** | **Fru17** (Frutos-Pascual & García Zapirain, 2017) | Review of the Use of AI Techniques in Serious Games: Decision Making and Machine Learning | 129 | 2005–2014 |
| | | **Abd22** (Abd-alrazaq et al., 2022) | Artificial Intelligence–Driven Serious Games in Health Care: Scoping Review | 46 | 2010–2021 |
| | **Games** | **Kha21** (Khakpour & Colomo-Palacios, 2021) | Convergence of Gamification and Machine Learning: A Systematic Literature Review | 43 | 2014–2019 |
| | | **Lu22** (Lu & Li, 2022) | Techniques and Paradigms in Modern Game AI Systems | – | – |
| | **Adaptive SG** | **Sri23** (Srimathi & Anitha, 2023) | Application of artificial intelligence in adaptation of gamification in education: A literature review | 29 | 2017–2022 |
| **Focus on adaptivity and personalisation** | **Adaptive SG** | **Saj22** (Sajjadi et al., 2022) | Individualization in serious games: A systematic review of the literature on the aspects of the players to adapt to | 37 | 2008–2019 |
| | | **Lop22** (Lopes & Lopes, 2022) | A Review of Dynamic Difficulty Adjustment Methods for Serious Games | 12 | 2009–2021 |
| | | **Ayd23** (Aydin et al., 2023) | Examination of adaptation components in serious games: a systematic review study | 26 | 2000–2021 |
| | | **Tou23** (Toukiloglou & Xinogalos, 2023) | A Systematic Literature Review on Adaptive Supports in Serious Games for Programming | 18 | 2000–2022 |
| | **Adaptive Games** | **Zoh18** (Zohaib, 2018) | Dynamic Difficulty Adjustment (DDA) in Computer Games: A Review | 84 | 2009–2018 |
| | | **Dal21** (Dalponte Ayastuy et al., 2021) | Adaptive gamification in Collaborative systems, a systematic mapping study | 25 | 2009–2019 |
| | | **Nin21** (Ninaus & Nebel, 2021) | A Systematic Literature Review of Analytics for Adaptivity Within Educational Video Games | 10 | 2012–2020 |
| | | **Ben22** (Bennani et al., 2022) | Adaptive gamification in E-learning: A literature review and future challenges | 23 | 2011–2021 |
| | | **Par23** (Paraschos & Koulouriotis, 2023) | Game Difficulty Adaptation and Experience Personalisation: A Literature Review | 109 | 2005–2021 |

N = Number of articles reviewed.

**Table 2**
List of research articles considered for the scoping review.

| Topic | | ID | Title | Algorithm(s) |
|---|---|---|---|---|
| | **Adaptive SG** | **BOU23** (Bouatrous et al., 2023) | A new adaptive VR-based exergame for hand rehabilitation after stroke | K-means |
| | | **ESF19** (Esfahlani et al., 2019) | Fusion of Artificial Intelligence in Neuro-Rehabilitation Video Games | FL, ANN |
| | | **EUN23** (Eun et al., 2023) | Artificial intelligence-based personalised serious game for enhancing the physical and cognitive abilities of the elderly | Rule-based learning |
| | | **GHO22** (Ghorbani et al., 2023) | Towards an Intelligent Assistive System Based on Augmented Reality and Serious Games | FL |
| | | **IZO21** (Izountar et al., 2021) | Towards an adaptive Virtual Reality Serious Game System for Motor Rehabilitation based on Facial Emotion Recognition | CNN |
| | | **KAL23** (Kalafatis et al. et al., 2023) | Artificial Intelligence Based Procedural Content Generation in Serious Games for Health: The Case of Childhood Obesity | GA |
| | | **KEM22** (Kempitiya et al., 2022) | Personalised Physiotherapy Rehabilitation using Artificial Intelligence and Virtual Reality Gaming | Hyperseed, MLP, and Machine Theory of Mind |
| | | **KOS23** (Kostkova et al., 2023) | Dynamic Difficulty Adjustment in Serious Games for Cerebral Visual Impairment | RL |
| | | **STR22** (Streicher & Aydinbas, 2022) | Bayesian Cognitive State Modelling for Adaptive Serious Games | NB |
| | | **CHE24** (Chen et al., 2024) | Generative AI-Based Difficulty Level Design of Serious Games for Stroke Rehabilitation | Generative Adversarial Network |
| | | **PEL24** (Pelosi et al., 2024) | Personalised rehabilitation approach for reaching movement using reinforcement learning | RL |
| **Multimodal data** | | **DIA22** (Dias et al., 2022) | MultiGRehab: Developing a Multimodal Biosignals Acquisition and Analysis Framework for Personalizing Stroke and Cardiac Rehabilitation based on Adaptive Serious Games | CNN |
| | | **DIA23** (Dias et al., 2023) | Multisensed Emotions as Adaptation Controllers in Human-to-Serious NeuroGames Communication | CNN |
| | | **MIT22** (Mitsis et al., 2022) | A Multimodal Approach for Real-Time Recognition of Engagement towards Adaptive Serious Games for Health | **No AI applied** |
| | **SG for diagnosis** | **ALJ19** (Aljumaili et al., 2019) | Serious Games and ML for Detecting MCI | RL |
| | | **GOU21** (Goumopoulos et al., 2021) | Applying Serious Games and Machine Learning for Cognitive Training and Screening: the COG-NIPLAT Approach | Comparison of: GNB, DT, SVM (best), k-NN, LR |
| | | **HOU23** (Hou et al.) | Analysis of Heart Rate Variability and Game Performance in Normal and Cognitively Impaired Elderly Subjects Using Serious Games | DTs, generated with a Tree-based Pipeline Optimisation Tool (TPOT) |
| | | **KAR21** (Karapapas & Goumopoulos, 2021) | Mild Cognitive Impairment Detection Using Machine Learning Models Trained on Data Collected from Serious Games | Comparison of: SVM (best), GNB (best), DT, k-NN, RF, LR, MLP |
| | | **KIM21** (Kim, An, & Park, 2021) | A Prediction Model for Detecting Developmental Disabilities in Preschool-Age Children Through Digital Biomarker-Driven Deep Learning in Serious Games: Development Study | DNN |
| | | **MAH22** (Mahboobeh et al., 2022) | Machine Learning-Based Analysis of Digital Movement Assessment and ExerGame Scores for Parkinson's Disease Severity Estimation | Comparison of: SVM, k-NN (best), RF |
| | | **MEZ22** (Mezrar & Bendella, 2022) | Machine learning and Serious Game for the Early Diagnosis of Alzheimer's Disease | Comparison of: NB, DT, RF (best), Adaboost, ExtraTree, LR |
| | | **NAI23** (Nair & Sakthivel, 2023) | A Deep Learning-Based Upper Limb Rehabilitation Exercise Status Identification System | Proposal: Deep CNN (best); Comparison with NB, DT, SVM, LD |
| | | **PIA23** (Piazzalunga et al., 2023) | Investigating Visual Perception Impairments through Serious Games and Eye Tracking to Anticipate Handwriting Difficulties | Proposal: CatBoost (best); Comparison with SVM, RF, NB |
| | | **TOK23A** (Toki et al., 2023a) | Applying Neural Networks on Biometric Datasets for Screening Speech and Language Deficiencies in Child Communication | Comparison of: SVM, kNN, ANN, GAs, PSO, ADAM, BFGS, INN (best) |
| | | **TOK23E** (Toki et al., 2023b) | Employing Classification Techniques on SmartSpeech Biometric Data towards Identification of Neurodevelopmental Disorders | DNN |
| | | **M23** (M & Surendran, 2023) | Chess Game to Improve the Mental Ability of Alzheimer's Patients using A3C | RL (Asynchronous Advantage Actor Critic) |
| **Multimodal data** | | **ESF22** (Esfahlani et al., 2022) | Machine Learning role in clinical decision-making: Neuro-rehabilitation video game | Comparison of: k-NN (best), SVM |
| | | **UPA23** (Upadhyay, 2023) | MemSpark: An Artificially Intelligent Virtual Reality System for Non-Intrusive Cognition Therapy and Evaluation of Dementia | MLP |
| | | **LLO24** (Llorente et al., 2024) | Assessment of cognitive games to improve the quality of life of Parkinson's and Alzheimer's patients | NB |
| | | **ABI24** (Abirami et al., 2024) | DeepMind Trainer: Harnessing Cognitive Neural Networks on Raspberry Pi for Advanced Cognitive Enhancement Games | CNN |
| | | **ALS25** (Alsheikhy et al., 2025) | Developing machine learning models for personalised game-based stroke rehabilitation therapy in virtual reality | RNN(Bi-LSTMs) and FFO |
| | **SG** | **BOU19** (Boulton et al., 2019, pp. 3921–3927) | Mobile Developments to Support Learners in Mainstream Education | Supervised learning (not explicitly stated) |
| | | **GIA19** (Giannakos et al., 2019) | Multimodal data as a means to understand the learning experience | RF |
| | | **HEN20** (Henderson et al., 2020) | Improving Affect Detection in Game-Based Learning with Multimodal Data Fusion | Comparison of: NB, SVM, RF, LR, MLP (best) |
| | | **MAT21** (Mat Sanusi et al., 2021) | Table Tennis Tutor: Forehand Strokes Classification Based on Multimodal Data and Neural Networks | RNN |

(continued on next page)

**Table 2** (*continued*)

| Topic | ID | Title | Algorithm(s) |
|---|---|---|---|
| | **SHA22** (Sharma et al., 2022) | Keep Calm and Do Not Carry-Forward: Toward Sensor-Data Driven AI Agent to Enhance Human Learning | Weighted average of seven ML algorithms |
| | **SVO20** (Svoren et al.) | Toadstool: A Dataset for Training Emotional Intelligent Machines Playing Super Mario Bros | CNN |
| | **JIA23** (Jiang et al., 2023) | A Serious Game System for Upper Limb Motor Function Assessment of Hemiparetic Stroke Patients | Proposal: FL (best); Comparison with SVM, k-NN, RT |
| | **MAS23** (Maskeliunas et al., 2022) | Deep Reinforcement Learning-Based iTrain Serious Game for Caregivers Dealing with Post-Stroke Patients | DRL |
| | **NAS20** (Nasri et al., 2020) | An sEMG-Controlled 3D Game for Rehabilitation Therapies: Real-Time Time Hand Gesture Recognition Using Deep Learning Techniques | Deep CNN |

critical issues. The most relevant outcomes were that research in this field had significantly risen during those years, with the highest number of publications in 2020. Here, most publications took place in the health sector (medicine (42) and health professions (15), considering overlaps), and the most common health conditions addressed were in this order: mental health, physical rehabilitation, chronic diseases, nutrition, and drug prevention. The most common game mechanics and design elements used in SGs and Gamification for health are in this order: Virtual Reality (VR) and immersion, storytelling, coaching, assessment, social collaboration, points/badges/quests/virtual rewards.

The research gaps identified by Dam23 (Damaševičius et al., 2023) include a lack of standardisation, insufficient user evaluation, missing long-term studies, unclear optimal dosage and duration, limited research on multiplayer game platforms, and sustaining player interest in healthy behaviours. There is also limited understanding of the mechanisms that lead to positive health outcomes, how to integrate SGs into healthcare systems, and effective game design for health behaviours. The authors recommend that future research should enhance user engagement, tailor games to specific patient needs, promote healthy behaviours, rigorously evaluate SGs, and address ethical and privacy concerns.

The enhancement of user engagement and the tailoring of the games for specific needs were our main concerns and led to our analysis of the remaining articles. We wanted to find out how and in which parts games have already been adapted, with or without the use of AI, and we aimed to draw more concrete conclusions about which methods and algorithms are useful and where exactly the research should go.

*3.2.1. Reviews with a focus on AI*

Fru17 (Frutos-Pascual & García Zapirain, 2017) reviewed 129 studies on AI techniques used in serious games between 2005 and 2014, splitting them into 88 studies focused on decision-making and 41 on machine learning (ML) algorithms. It is a comprehensive review guided by the first three flagships mentioned by Yannakakis (Yannakakis, 2012). The authors grouped the research contributions by AI techniques and analysed usage and types of games for each flagship. We inferred from their narrative text the frequency of usage for game control and user assessment tasks and added it to Table 3.

- For game control, Decision Trees (DT) (16 studies), Finite State Machines (FSM) (14 studies), and Rule-Based Systems (4 studies) have mostly been applied to influence both gameplay and the behaviour of Non-Player Characters (NPCs). Fuzzy Logic (FL) (10 studies), Goal-Oriented Action Planning (GOAP) (6 studies), Artificial Neural Networks (ANN) (7 studies), and Support Vector Machines (SVM) (1 study) were used to control NPC behaviour, while Case-Based Reasoning (CBR) (8 studies), Hidden Markov Models (HMM) (7 studies), and Naïve Bayes (NB) (2 studies) were applied to adapt the gameplay.
- The algorithms related to user assessment included NB (11 studies), SVM (5 studies), GOAP (2 studies), and CBR (1 study) to analyse the user's behaviour and interaction with the game; DT (8 studies) and HMM (3 studies) to analyse user motivation; Rule-Based Systems (7 studies) to assess performance and analyse facial expressions; as well as FL (5 studies), ANN (5 studies), and FSM (3 studies) to classify the user's state.

Abd22 (Abd-alrazaq et al., 2022) conducted a scoping review on the use of AI-driven SGs in healthcare, analysing 46 studies (25 journal articles and 21 conference papers) published between 2010 and 2021. The review states no clear research questions despite having followed the PRISMA-ScR checklist. Results are presented in narrative form, summarising different characteristics of the games and AI techniques by providing the statistics about the targeted health conditions, purposes, modalities, platforms, data types, etc. The most interesting conclusion for our research is their analysis of AI Techniques: 89 % of the included studies used algorithms to solve classification problems, 11 % regression problems, and 4 % clustering problems. 27 different algorithms were mentioned here, the most frequently used were: SVM (30 %), CNN (15 %), ANN (15 %) and RF (15 %). 9 different purposes have been detected: detection of disease (28 %), evaluation of user performance (28 %), adaptation of difficulty level (15 %), recognition of gestures (15 %), recognition of biosignals (11 %), supporting users to play (7 %), classification of activity (4 %), recognition of voice (4 %), and prediction of user characteristics (2 %). These are almost all related to user assessment; only 4 works used AI for game control mechanisms (classified by the authors as "adaptation of difficulty level"). Unfortunately, the authors did not provide sufficient deep information needed for our research to classify the used AI algorithms according to our scheme. The tables provided as multimedia appendices revealed some more details, but for many of the articles, multiple AI methods were listed, and it was not clear which of them were used for comparison and which were proposed. As a consequence, we needed to revise 14 articles to find out which algorithms were considered most appropriate for the desired task, to add them to Table 3. Studies that tested more than one algorithm have been added multiple times. Our analysis revealed the following.

- Five studies used AI for game control mechanisms, all aimed at adjusting difficulty levels according to the user's performance. The algorithms used were FL (3 studies), ANN (2 studies), and Reinforcement Learning (RL) (1 study); one study used both FL and ANN.
- The remaining studies focused on user assessment with various algorithms. SVM was used in 15 studies for tasks including disease detection (7 times), gesture classification (4 times), performance evaluation (3 times), and biosignal recognition (1 time). ANN appeared in 9 studies, primarily for gesture classification and disease detection. K-Nearest Neighbours (k-NN) was found in 8 studies, evenly split between disease detection and gesture recognition. DT was mentioned 7 times, mainly for disease detection. Convolutional Neural Network (CNN) and Random Forest (RF) were each used in 6 studies for user performance, gesture recognition, and disease detection. FL and NB were applied in a few studies for performance evaluation, disease detection, and activity classification. Additionally, k-means clustering, AdaBoost, and Regression Tree (RT) were each used in isolated studies for similar tasks.

The following reviews do not focus explicitly on SGs but still on the usage of AI.

Kha21 (Khakpour & Colomo-Palacios, 2021) examined 43 articles from 2014 to 2019, focusing on the relationship between gamification and machine learning (ML) algorithms—specifically, how ML is applied in gamification and vice versa. Of these, 20 studies were analysed based on the algorithms used: 6 applied AI algorithms to game control tasks, while 15 focused on assessing the user's physical state.

- Bayesian Network were employed in two studies for adapting game difficulty. Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) were each used in one study for developing intelligent game elements. The CN2 induction algorithm was applied in two studies for selecting a personalisation strategy, and ε-greedy variations were also utilised in two studies for game content adaptations.
- For user assessment, the most frequently employed technique was Support Vector Machine (SVM), used in four studies to predict the user's affective state, performance, and to classify game-specific behaviours. DTs were used in three studies: one study combined them with k-NN and kernel optimisation to assess user performance; another combined a DT with a Markov process for finger recognition; and a third applied DTs for disease classification by assessing the knowledge level. Additionally, various other algorithms were reported in individual studies: expectation-maximization clustering for classifying performance, k-means for assessing brain status, Artificial Neural Networks (ANN) for evaluating driving behaviour, Logistic Regression (LR) for modelling past user behaviour, and an ML agent to build a model estimating reading knowledge. Two studies used Hidden Markov Models (HMM) and Random Forests (RF) to monitor player activity.

However, the review does not provide an in-depth analysis of these techniques, leaving unclear which algorithms are most suitable or most frequently used for specific tasks. The only standout algorithm is SVM (as found by Abd22 (Abd-alrazaq et al., 2022)), which appears well-suited for predicting user performance and affective state. Nonetheless, we have included these findings in Table 3 as referenced.

Lu22 (Lu & Li, 2022) provided a comprehensive review of AI techniques in games, relevant to our RQ1, but it lacks the rigour of a systematic review, omitting key details like the number of articles reviewed and the period covered. The algorithms used in each study were not specified, preventing inclusion in our table. Despite this, the review's focus on game control algorithms, particularly those categorised as "decision intelligence," is valuable. These methods combine knowledge with real-time planning and reasoning, involving techniques like Minimax, Monte Carlo Tree Search, and continual resolving. Learning often utilises genetic algorithms, supervised learning, Reinforcement Learning (RL), or Multi-Agent Learning. Deep reinforcement learning (DRL) is common in complex games. Single-agent games are generally modelled as Markov Decision Processes with RL, while supervised learning in modern game AI systems often uses ANNs, including CNNs and RNNs for spatial and temporal feature extraction.

The last study reviewed in this section is Sri23 (Srimathi & Anitha, 2023), which focused on AI applied to the personalisation of gamified environments. The authors examined 29 studies on the use of various gamification elements, user profile components, and intelligent techniques. They analysed how different gamification elements, which they allocated to three categories according to frequency of usage, are processed with AI techniques. The most used gamification elements were identified as badges, leader boards, levels, points and progress bars. The purposes of the algorithms were not exactly indicated, but we derived from the text the following categorisation for our classification.

- 11 algorithms were used for game control-related tasks: One study reports a model that combines logistic regression, NB, SVM, RF, and ANN. RL was used by two studies, FL by one, to suggest an adaptive learning path. DT was used by one study, and another study compared ANN and Adaptive Neuro-Fuzzy Inference System (ANFIS).
- Two algorithms were identified for user assessment: FL for evaluating player access to the platform and a two-step clustering algorithm for assessing the use of gamification elements and academic performance.

However, extracting further detailed information was challenging, as some algorithms were either not identified or insufficiently described.

### 3.2.2. Reviews with a focus on personalisation and adaptation

Having reviewed various sources focused on AI, we then turned to studies that addressed the personalisation and adaptation of games and serious games (SGs) without explicitly mentioning AI. We aimed to identify the tasks involved in (Table 3) this process and the input data used, specifically how users were assessed, and what the adaptation criteria were. This analysis may reveal AI algorithms in use or suggest potential AI techniques that could enhance these processes. Additionally, the discussion sections of these reviews might offer insights into possible AI applications.

As in the previous section, we first study reviews about SGs & exergames (Saj22 (Sajjadi et al., 2022), Lop22 (Lopes & Lopes, 2022), Ayd23 (Aydin et al., 2023) and Tou23 (Toukiloglou & Xinogalos, 2023)), later we also contemplate articles about games in general (Zoh18 (Zohaib, 2018), Dal21 (Dalponte Ayastuy et al., 2021), Nin21 (Ninaus & Nebel, 2021), Ben22 (Bennani et al., 2022), and Par23 (Paraschos & Koulouriotis, 2023)).

Saj22 (Sajjadi et al., 2022) reviewed 37 articles from 2008 to 2019 to know which player aspects serious games (SGs) should adapt to. It found out that player performance aspects, such as in-game skills, were the most frequently researched for individualization, while physiological states and personal traits were less studied. Personalisation based on performance commonly involves adjusting difficulty, levels, and feedback. Of the 21 works cited, one used an undefined ML algorithm, one used a Hidden Markov Model (HMM), and one used Fuzzy Logic (FL). Thirteen works mentioned adaptation based on physiological parameters and affective states without specifying algorithms, while 12 studies discussed personal traits without noting AI applications. The authors observed that ML methods for profiling players were not widely used in adaptive serious gaming but noted a growing trend. Unfortunately, the article lacked the specific details needed to include their results in Table 3.

Lop22 (Lopes & Lopes, 2022) reviewed dynamic difficulty adjustment (DDA) methods in serious games and entertainment games, analysing 12 studies from 2009 to 2021–5 on entertainment games, 5 on motor rehabilitation games, and 2 on mental rehabilitation games. The review aimed to identify strategies and techniques for adapting game difficulty to enhance player performance, engagement, and satisfaction.

- From their narrative description, we deduced that in terms of game control, Reinforcement Learning (RL) was most frequently used to adapt to the difficulty of the game (3 studies), often dynamically during the game, depending on the player's performance and skills. The advantage of RL is its ability to learn during the game without prior training of the model, but the employment of FL to adapt the level of difficulty was also mentioned. To modify the game scenario, one study used deep RL and another deep NN. Monte Carlo Tree Search was mentioned as useful to increase the number of tasks and difficulty.
- Concerning user assessment, 2 studies applied Deep NN to determine the level of anxiety. One paper used RL to model user behaviour and one study used a Regression Tree (RT) to model the physiological state of the player.

**Table 3**
Number of AI algorithms found in each review article and in the scoping review (Tao25), ordered by the year of publication, classified according to the type of usage: game-oriented (G) or user-oriented (U).

| | | Fru17 | | Zoh18 | | Kha21 | | Abd22 | | Lop22 | | Ayd23 | | Sri23 | | Tou23 | | Par23 | | Tao25 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | G | U | G | U | G | U | G | U | G | U | G | U | G | U | G | U | G | U | G | U |
| **Cat.** | **Class** | | | | | | | | | | | | | | | | | | | | |
| **SL** | **Decision Trees (gen.)** | 16 | 8 | | | | 3 | | 7 | | | | 1 | 1 | | | | | | | 1 |
| | Random Forest | | | | | | 1 | | 6 | 1 | | | | 1 | | | | | 1 | | 2 |
| | Ada Boost | | | | | | | | 1 | | | | | | | | | | | | |
| | Regression Tree | | | | | | | | 1 | | 1 | | | | | | | | 1 | | |
| | Cat Boost | | | | | | | | | | | | | | | | | | | | 1 |
| | **Rule-Based Systems (gen.)** | 4 | 7 | | | | | | | | | | | | | 1 | | | | 1 | |
| | CN2 | | | | | 1 | | | | | | | | | | | | | | | |
| | Simple Adaptation | | | | | | | | | | | 4 | | | | | | | | | |
| | **Linear classifiers** | | | | | | | | | | | | | | | | | | | | |
| | Logistic Regression | | | | | | 1 | | | | | | | 1 | | | | | 3 | | |
| | Support Vector Machine | 1 | 5 | | | | 4 | | 15 | | | | | 1 | | | | 1 | 1 | | 2 |
| | **Nonlinear classifiers** | | | | | | | | | | | | | | | | | | | | |
| | Naïve Bayes | 2 | 11 | 1 | | 1 | | | 2 | | 1 | 1 | 3 | 1 | | 3 | | 1 | | 1 | |
| | Hidden Markov Model | 7 | 3 | | 1 | | 2 | | | 1 | | | | | | | | 1 | 1 | | 2 |
| | **Fuzzy Logic (gen.)** | 10 | 5 | 1 | | | | 3 | 3 | 1 | | 3 | 1 | 1 | 1 | | 4 | | 1 | 2 | 1 |
| | ANFIS | | | | | | | | | | | | | 1 | | | | | | | |
| | **Artificial Neural Networks (gen.)** | 7 | 5 | 5 | 2 | | 1 | 2 | 9 | | | | 1 | 2 | | | | 2 | 1 | | 1 |
| | Multilayer Perceptron | | | 3 | | | | | | | | | | | | | | | 1 | | 2 |
| | Integer bounded NN | | | | | | | | | | | | | | | | | | | | 1 |
| | Recurrent NN | | | | | 1 | | | | | | | | | | | | | | | 2 |
| | Convolutional NN (also deep) | | | | | 1 | | | 6 | | | | | | | | | | | 2 | 4 |
| | Deep Learning NN | | | | | | | | | 1 | 2 | | 1 | | | | | | 1 | | 2 |
| | Finite State Machines | 14 | 3 | | | | | | | | | | | | | | | | | | |
| | Kernel methods (k-NN) | | | | | | 1 | | 8 | | | | | | | | | 1 | | | 2 |
| **UL** | k-means Clustering | | | | | | 1 | | 2 | | | | | | | | | | | 1 | |
| | Expectation-Maximization | | | | | | 1 | | | | | | | | | | | | | | |
| | Two-step Clustering | | | | | | | | | | | | | | 1 | | | | | | |
| | Hyperseed | | | | | | | | | | | | | | | | | | | | 1 |
| **RL** | **Reinforcement ML (gen.)** | | | 4 | | | | 1 | | 3 | 1 | | | 2 | | | | 3 | | 2 | 2 |
| | Monte Carlo Tree Search | | | | | | | | | 1 | | | | | | | | | | | |
| | Deep Reinforcement Learning | | | | | | | | | 1 | | | | | | | | | | 1 | |
| | ε-greedy | | | | | 1 | | | | | | | | | | | | | | | |
| **O** | Goal Oriented Behaviour | 6 | 2 | 1 | | | | | | | | | | | | | | | | | |
| | Case-Based Reasoning | 8 | 1 | | | | | | | | | | | | | | | | | | |
| | Information Field Theory | | | | | | | | | | | 2 | | | | | | | | | |
| | Item Response Theory | | | | | | | | | | | 2 | | | | | | | | | |
| | Hamlet | | | 1 | | | | | | | | | | | | | | | | | |
| | Dynamic Scripting | | | 1 | | | | | | | | | | | | | | 1 | | | |
| | Neuro-evolution | | | 4 | | | | | | | | | | | | | | 1 | | | |
| | Minimax alg., alfa-beta pruning | | | | | | | | | | | | | | | | | 1 | | | |
| | Swarm intelligence | | | | | | | | | | | | | | | | | 1 | | | |
| | Graph optimisation | | | | 1 | | | | | | | | | | | | | | | | |
| | Genetic Algorithms | | | | | | | | | | | | | | | | | | | 2 | |
| | Machine Theory of Mind | | | | | | | | | | | | | | | | | | | 1 | |

Abbreviations: SL-Supervised Learning, UL-Unsupervised Learning, RL-Reinforcement Learning, O-Others, NN-Neural Network, gen.-generic.

Ayd23 (Aydin et al., 2023) also investigated adaptive components in SGs, reviewing 26 articles from 2000 to 2021 on adapting educational games through modifications to the game environment, educational content, interface, or NPC behaviour using a user model. The topic has a narrower focus than Fru17 (Goršič et al., 2017) but being a systematic review and presenting a wider period, it was expected to find some overlap in the results. Nevertheless, the outcomes are very different and none of the articles is cited in both reviews. Only 19 studies employed AI and were classified by algorithm type, which we included in Table 3. Game elements were categorised under game control, while factors for user model creation were classified as user assessment. To determine which decision systems were used for specific purposes, we reviewed the referenced articles.

- The review identified various decision systems applied to adjust the game content depending on the user's state or performance. Rule-based adaptation, using simple rules, was the most commonly used technique to adapt the game according to the player's state and provide feedback (4 studies). Three works employed FL as an advice generator, to control the game level and to personalise the game content. Additionally, Item Response Theory was used to adapt the difficulty according to the user's responses (2 studies) and Information Field Theory (1 study) to adapt the content of the game. One work was cited using an NB to adapt the game elements, the content, and the NPC behaviour.
- In terms of user assessment, traditional techniques were prevalent, with NB being the most used (3 studies) to build a player's model. Two studies applied ANN, one of them also Deep Learning. One used a DT and one FL to represent the learner's knowledge level. Adaptation methods varied: static adaptation used pre-game data, while dynamic adaptation was based on in-game data, such as player performance and behaviour.

Tou23 (Toukiloglou & Xinogalos, 2023) is a scoping review focused on intelligent adaptive mechanics in SGs. It examined 18 studies published over 23 years (2000–2022).

- Three studies used NB for game control, training a user model to adjust content based on player progress, while one study employed a rule-based algorithm for similar purposes.

- Additionally, four studies applied FL for user assessment, concretely to represent the student knowledge level and cognitive state of the player. The advantage of FL is the uncertainty, such that the knowledge level could be described using partial levels instead of distinct values.

In the following, we analysed the articles with a more general focus on games and gamification.

The earliest review in our selection is Zoh18 (Zohaib, 2018), which analysed 84 studies from 2009 to 2018. Although its title is similar to that of Lop22 (Lopes & Lopes, 2022), which focused on Dynamic Difficulty Adjustment (DDA) for SGs, Zoh18 (Zohaib, 2018) specifically examined difficulty adaptation to keep players in an optimal flow channel—a concept introduced by R. Koster in 2004 (Koster, 2014). The review classified the DDA approaches found according to seven methods proposed in the literature (Probabilistic Methods, Single and multi-layered perceptron, Dynamic scripting, Hamlet System, Reinforcement Learning, Upper Confidence Bound for Trees and ANN, and Self-organizing System and ANN). We identified 25 studies from their text (one of which was also cited in Kha21 (Khakpour & Colomo-Palacios, 2021)), which we could classify according to the methods listed in Table 3.

- Twenty-one of the cited studies dealt with game control: five works applied an ANN, four of which adjusted the NPC behaviour (two of them applying a neuroevolutionary network) and three used a multilayer perceptron to change the gaming environment. Four works applied RL for difficulty adaptation, game content creation, and adjustment of NPC behaviour. The remaining works cited used various techniques: Dynamic Scripting and HMM to control the NPC behaviour (opponents), Goal-Oriented Action Planning (GOAP) and the Hamlet System to adjust the gaming environment through economic rules, as well as fuzzy rules to adapt game tactics.
- Four works focused on the creation of different types of user models to make decisions to adjust the game difficulty, so we assigned these techniques to user assessment. Two of them applied ANN, both modelling the players' emotion, e.g. to distinguish fun, frustration and challenge., Furthermore, one approach modelled the progression of the player using graph optimisation, and one study found how to estimate the player's behaviour and personality by applying an HMM.

The two reviews following Zoh18 (Zohaib, 2018) were Dal21 (Dalponte Ayastuy et al., 2021) and Nin21 (Ninaus & Nebel, 2021). Dal21 did not address AI, Nin21 did not provide technical details on adaptation methods; thus, neither was offering useful data for Table 3. However, their findings align with those discussed before, so we summarise this information in the following paragraphs.

Dal21 (Dalponte Ayastuy et al., 2021) reviewed 25 articles on gamification in collaborative systems published between 2009 and 2019. They found that most studies focused on difficulty adaptation, primarily based on performance measurement, with few addressing storytelling adaptations. The review highlighted those 18 studies used Procedural Content Generation (PCG) for managing difficulty levels, 10 altered the behaviour of the agent, and 9 implemented motivational interventions.

Nin21 (Ninaus & Nebel, 2021) examined 10 articles on adaptivity in educational video games from 2012 to 2020. Of these, four utilised AI algorithms: cognitive load theory, a Bayesian network for estimating user knowledge, competence-based knowledge space theory for agents, and modular Reinforcement Learning. The authors noted that the reviewed articles often lacked detailed descriptions of the games and adaptive mechanisms used.

Ben22 (Bennani et al., 2022) reviewed articles from 2011 to 2021 on adaptive gamification in online education, focusing on the role of AI. Although the review included an overview of 13 studies, with 9 using AI algorithms, insufficient detail was provided for our results table. Nevertheless, the article highlights significant potential for enhancing gamification with AI advancements. Seven studies indicated the use of machine learning (ML) without specifying the techniques. Two studies applied Case-Based Reasoning (CBR), one of them combining it with ANN. Most studies used these algorithms to adapt to the gamification environment, categorised under game control, while one predicted the learner's initial ability.

Finally, Par23 (Paraschos & Koulouriotis, 2023) provided an extensive overview of research on game difficulty adaptation and personalisation from the perspectives of DDA and PCG. By reviewing 109 articles from 2005 to 2021, they identified the most relevant studies starting from 2015. However, since DDA does not necessarily require AI, their focus was not on these algorithms. Nonetheless, many studies used AI to create user models, control agents, or manage game aspects. Par23 (Paraschos & Koulouriotis, 2023) categorised the reviewed works into four methods of game adaptation: (a) Performance-based adaptation, (b) Affective-based adaptation, (c) Player-experience modelling, and (d) Agents. This categorisation does not completely align with ours, and as the authors provided limited information about the AI algorithms applied and their specific purposes, we searched for the original publications in some cases to obtain more details. Ultimately, we were able to categorize 26 works as follows.

- For game control: Four works applied RL for different purposes, such as balancing game environments, creating a difficulty ranking system, and controlling NPC behaviour. ANN was mentioned in two works to adjust the scenario or difficulty according to the player's performance. Furthermore, nine works applied various algorithms to control NPC behaviour and difficulty, including NB (2 works), dynamic scripting, HMM, k-NN, neuro-evolution ANN, swarm intelligence, and the minimax algorithm with alpha-beta pruning. The game space was personalised in one work using RF, and in another work using an SVM to tailor a racing circuit.
- Regarding player assessment: Linear regression was mentioned three times as a method to adjust game elements and activities based on the player's physiological data, performance, or responses to questionnaires. The remaining works used different algorithms. Four evaluated physiological data using deep learning, regression trees, multilayer perceptron, and SVM. Three measured the player's interactions, fitting them into HMM, ANN, or RF classifiers; one used RL on the performance metrics; and one used FL to estimate the player's knowledge level.

At this stage, Table 3 presents the results from 9 reviews, ordered chronologically to show the usage of each algorithm over time. For example, Decision Trees (DT), Naive Bayes (NB), and Finite State Machines (FSM) were used frequently in earlier years but have nearly disappeared. In contrast, Fuzzy Logic (FL) has remained relatively constant, while Artificial Neural Networks (ANN) have been consistently used but have evolved into more specific implementations such as Convolutional Neural Networks (CNN), Iterative Neural Networks (INN), and deep learning. Fig. 4 visualises the total usage of each algorithm, facilitating a comparison between their application to game control (blue) and user assessment (orange). Additionally, some percentages have been added to the most frequently applied algorithms to see the difference between usage for game control and user assessment.

### 3.3. Scoping review of new research proposals

This section continues from the meta-review to assess whether recent publications align with previously used algorithms or introduce new approaches. In the following two subsections, the articles are presented according to game control-centred and user-centred applications; within each section, they are organised by algorithms, in the same order as listed in Table 3 and added to it in the final column named Tao25. The

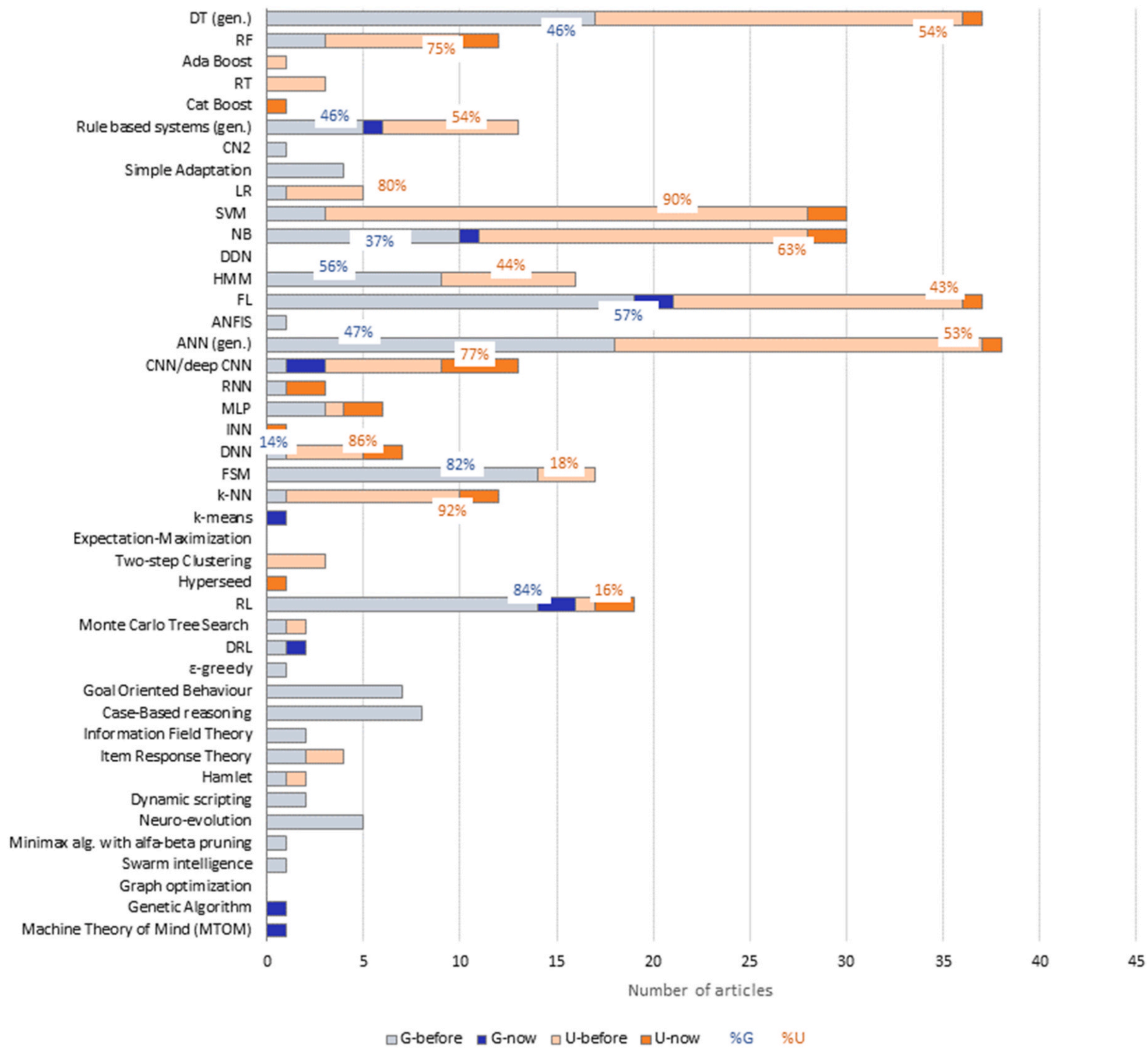


**Fig. 4.** Number of AI algorithms found to be used for game control (blue) and user assessment (orange). Light-coloured parts correspond to the algorithms found in the meta-review, and dark-coloured parts to the scoping review (G-new and U-new). Some percentages have been added to compare the usages for game control or user assessment for the most frequently used algorithms. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

third subsection revises the works dealing with multimodal data.

Works that compare different algorithms are cited multiple times in different sections, highlighting the performance and suitability of each algorithm within those studies. When a work is cited for the first time, details about the study purpose and types of patients are given but not repeated in the following cites.

*3.3.1. Game control-centred studies*

In the case of game control-centred applications, the main task of the AI algorithm is to configure and readjust the gaming environment and/or difficulty to enhance the success of therapy. Decisions are primarily based on information gathered from players, which is usually their performance. However, emotional and physiological data are also frequently considered, as they offer insights into the player's motivation, which should ideally be maximized to achieve better results.

Other interesting input data could be the health status, rehabilitation goals, rehabilitation history, personal preferences, and game progress: if multiple of these data are applied, the method is classified as multimodal data. The AI could be used for developing an appropriate rehabilitation program, executing a specific exercise program, controlling exercise intensity, as well as determining the frequency of rehabilitation activities and rest cycles, thus replacing the traditional decision-making process carried out by the therapist.

a) Decision Tree (DT)

A DT, introduced by J. R. Quinlan in 1986 (Quinlan, 1886), is a supervised learning algorithm utilised for classification and prediction. The meta-review revealed frequent usage for game control, also variations like Regression Tree (RT), Random Forest (RF), Ada Boost and Cat Boost were applied, but now recent works have been found that used any kind of DT for game control.

b) Rule-Based Systems

Rule-based learning is an ML approach in which a model's decisions are based on a predefined set of rules, often used for domain-specific problems with explicit rules. In this approach, rules are usually defined manually by domain experts rather than being acquired through a data-driven learning process.

- EUN23 (Eun et al., 2023) developed a serious game designed to improve the physical and cognitive functions of elderly patients using gesture and body motion as input. The rule-based learning system sets an initial difficulty level based on each user's capabilities, with guidelines established by a group of fitness instructors. The game analyses the user's performance over five plays to determine the initial level and readjusts the difficulty during gameplay based on performance. If the average success rate is between 80 % and 90 %, the difficulty remains the same. If it exceeds 90 %, the difficulty increases by one level, and if it is below 80 %, the difficulty decreases by one level.

c) Linear Classifiers

In the category of linear classifiers, Logistic Regression (LR) is a linear model used to solve classification problems, usually for binary classification problems, and performs well in terms of computational speed and when dealing with large data sets. However, LR may not perform well for data with non-linear relationships as it assumes a linear relationship between features and targets. A Support Vector Machine (SVM) is an ML algorithm mainly used for classification and regression tasks, which has good generalisation ability and strong robustness, and performs well when dealing with high-dimensional and non-linear data.

The meta-review found some but few cases that applied Logistic Regression (LR) and Support Vector Machine (SVM) for game control. Here, no recent works have been found that used any kind of linear classifier.

d) Nonlinear Classifiers

Naïve Bayes (NB) and Hidden Markov Model (HMM) were the algorithms frequently found in the meta-review for this category. NB is a classification algorithm based on Bayes' theorem and the assumption of conditional independence of features. The NB classifier includes several variants such as Gaussian NB (GNB), Polynomial NB, Bernoulli NB, or Hierarchical Bayesian Modelling (HBM), each of which applies to different types of data and problems. One of the recent works used a variation of HBM (it was classified in Table 3 as NB).

- In STR22 (Streicher & Aydinbas, 2022), a serious game was designed to continuously motivate players by dynamically adapting the user experience. This was achieved using a user model based on Hierarchical Bayesian Modelling (HBM). The model estimates variables such as cognitive load and mental working memory capacity to gauge the players' mental states. By assessing these states, the game adjusts its difficulty and overall experience, enhancing engagement and satisfaction. The study results show that the proposed model accurately explains task success, task score, and task time, with over 90 % accuracy, demonstrating its effectiveness and feasibility in serious game design.

e) Fuzzy logic (FL)

FL is a heuristic approach that allows for more advanced decision-tree processing and better integration with rules-based programming, as it deals with reasoning that is approximate rather than fixed and exact. FL is a logical reasoning method that deals with fuzzy information and uncertainty. FL allows variables to be fuzzy or uncertain to some extent, taking values between 0 and 1. Multiple works were found in the meta-review that applied FL to game control. Here, two recent works have been identified.

- ESF19 (Esfahlani et al., 2019) utilised an FL reasoning scheme to recommend appropriate levels of application difficulty based on the player's performance and rehabilitation objectives. This approach aims to avoid unsuitable difficulty levels that could lead to further sports injuries, thereby enhancing rehabilitation outcomes and safeguarding the player's health and safety. The system was integrated with an artificial neural network (ANN) classifier that predicted improvement following the intervention, based on data collected from the user during play (including information from Kinect, Myo armband, and a foot pedal). We classified this component as user assessment and will provide a more detailed explanation in section 3.3.2 f).
- In GHO22 (Ghorbani et al., 2023), an adaptive FL algorithm is employed to recommend the number of tasks performed in a serious game designed for patients with mild cognitive impairment. The game simulates daily living situations and includes five tasks, such as finding a particular object, remembering colours, and arranging numbers. The input variables comprise various types of data from embedded devices, the user's real-time location, and their cognitive state. The output variables consist of different types of augmented reality (AR) messages (voice, image, and text), which are defined as fuzzy singleton membership functions. Each message is assigned an identification number and can be triggered by specific inputs. A triangular membership function is used to describe the states of the relay actuators.

f) Artificial Neural Networks (ANN)

Since ANN appeared, a great number of variations have been developed. In the meta-review, multiple applications of a generic ANN were found, as well as variations like Convolutional Neural Network (CNN), Recurrent NN, Multilayer Perceptron and Deep Learning NN. In the recent works, only three were identified using a CNN.

- In DIA22 (Dias et al., 2022) and DIA23 (Dias et al., 2023), the researchers decomposed multimodal bio signals by Swarm Decomposition and presented the data, separated into different frequencies, to a CNN. Through CNN processing, they identified and categorised the physical and mental stresses of the patient and fed the results as a feedback loop into a serious gaming rehabilitation scenario to adapt its difficulty level and intensity to the patient's current emotional state. In section 3.3.3 we will highlight the particularities of the multimodal data applied.
- In IZO21 (Izountar et al., 2021), the researchers utilised a CNN based on the Deep Face model proposed by Y. Taigman et al. (Taigman et al., 2014) to categorize the patient's emotions. They designed a self-adaptive system to determine the necessary changes based on the results obtained. After each detected change in the patient's mood, the system automatically adjusts the 3D exergame for motor rehabilitation, modifying the game's complexity, speed, and level according to the patient's emotional state.

g) Kernel Methods (k-NN)

Kernel methods are based on similarity learning. k-Nearest Neighbours (k-NN) is one that uses a metric feature space to compute distances to *k* data points and can be used for either classification or regression applications (in supervised learning) or anomaly detection (in unsupervised learning). It does not seem to be very helpful for game control as only one work was mentioned in one of the reviews and no recent work was found that used a kernel method for game control.

h) Unsupervised Learning

The k-means algorithm is an unsupervised learning algorithm that is widely used in fields such as data mining and pattern recognition to help discover hidden patterns and structures in datasets, as it does not require labelled training data, but instead clusters data samples based on their similarity to each other.

- In BOU23 (Bouatrous et al., 2023), the researchers used the k-means algorithm to perform a cluster analysis of the patient's performance during the game. The algorithm takes as input a list of previously recorded performances and classifies this data into three different clusters. The cluster containing the highest number of instances is selected as the gaming level, and the list of constraints is updated according to the selected cluster. The procedure includes a fatigue check during the game. The results showed a high level of confidence in the use of the system for 11 stroke patients.

i) Reinforcement Learning (RL)

RL is based on “agents” that learn to make decisions by interacting with an environment. By applying a trial-and-error strategy, the agent learns a policy that maximises cumulative rewards over time, according to the feedback received from the environment in the form of rewards or punishments. This method was frequently found applied to game control in the meta-review, as well as variations. Here, one work applied RL.

- In KOS23 (Kostkova et al., 2023), the authors implemented in-game adaptivity using a reinforcement learning approach to control the difficulty of a set of mini-games for children with cerebral visual impairment. They defined the states as descriptions of the game environment, actions as changes to game parameters according to the environment and the player’s performance, and rewards as feedback that indicates the quality of action for each of the four mini-games. The player’s performance was evaluated after each level, and the values were updated for each state-action pair.
- PEL24 (Pelosi et al., 2024) proposes a method that integrates VR and Reinforcement Learning (RL) to develop personalised rehabilitation programs for upper limb recovery. To address limitations of traditional physical therapy, such as lack of ecological validity, low patient engagement, and insufficient personalisation, the researchers designed a VR-based bubble-popping game. A Q-learning algorithm is employed to dynamically adjust the position of virtual bubbles based on the patient’s motor abilities. The algorithm learns from kinematic features—including trajectory length, smoothness, completion time, and peak velocity—and incorporates therapist-defined reward strategies to adapt game difficulty in real time. Simulation of two patient cases demonstrated the algorithm’s ability to adjust task distribution in response to changing motor performance. This work lays the groundwork for more flexible and adaptive VR rehabilitation systems, with future clinical trials planned to validate its effectiveness in real-world settings.

Deep Reinforcement Learning (DRL) applies, as the name suggests, deep learning techniques to an RL framework. In DRL, an intelligent body learns to perform a task by interacting with its environment, observing the state of the environment, acting, and receiving rewards.

- DRL was used in MAS23 (Maskeliunas et al., 2022) in a training game for caregivers of stroke patients. Depending on the player’s (caregiver) performance and knowledge level, the DRL was applied to control the game flow and content, i.e., choosing scenarios and gaming events from the set for each interaction. This helped improve caregivers’ knowledge and competence in post-stroke care, and good results were achieved.

j) Others

A genetic algorithm is a heuristic optimisation algorithm that has strong global search capability and robustness and does not depend on the particular structure of the problem.

- KAL23 (Kalafatis et al. et al., 2023) proposed a novel Genetic Algorithm-based approach for Personalised Content Generation (PCG) to provide customised messages and missions to fulfil in an SG aimed at combating childhood obesity, considering data collected from health-related sensors as well as the user’s interaction with the game.
- In KEM22 (Kempitiya et al., 2022), the researchers developed a framework for personalised physiotherapy rehabilitation using a VR game. This framework integrated an AI module with three layers: a) a profile of the patient’s exercise history was created using an unsupervised learning algorithm named Hyperseed; b) a Multilayer Perceptron (MLP) was employed to classify the current state of the patient; and c) a personalised game-level plan was determined based on the outcomes of a) and b). The first two components will be counted as user assessment in Table 3, the third component utilised dynamic difficulty adjustment and the machine version of the theory of mind (MTOM), applied as a mirror AI agent of the patient, to personalise the game level.
- In CHE24 (Chen et al., 2024), this study explores the use of generative artificial intelligence, specifically Generative Adversarial Networks (GANs), to design personalised difficulty levels for serious games aimed at stroke rehabilitation. Compared to traditional methods and Long Short-Term Memory (LSTM)-based approaches, GANs offer advantages in handling sequential data and adapting to dynamic demands, thereby streamlining the rehabilitation process and potentially reducing training time. The core research question addresses how to design individualised difficulty plans that align with patients’ rehabilitation stages and personal differences. To achieve this, demographic information such as the player’s age, height, weight, and severity of impairment is used as the input "noise" for the GAN model, while actual difficulty level data collected during gameplay serves as real data to train the network. The goal is to enable automated and adaptive generation of game difficulty levels, thereby enhancing the effectiveness of training and increasing patient engagement.

### 3.3.2. *Review of user-centred studies*

a) Decision Tree (DT)

DT algorithms and variations have been used in similar amounts for user assessment as for game control-related tasks. The meta-review reveals that in the earlier years, they were more frequently applied to game control (Fru17 (Frutos-Pascual & García Zapirain, 2017)), while in later years, the application to user assessment was prevalent (Abd22 (Abd-alrazaq et al., 2022)). In our revision of recent works, they have been found to solve the following problems.

- MEZ22 (Mezrar & Bendella, 2022) presented a pilot study about the early detection of mild cognitive impairment, comparing six different algorithms that processed the data obtained from a gamified cognitive tool. Four of them were trees (DT, Random Forest (RF), AdaBoost and Extra Trees), furthermore, they applied Naïve Bayes (NB) and Logistic Regression (LR). Here, RF, an ensemble learning algorithm capable of handling classification and regression tasks, achieved the best results. AdaBoost, which is usually able to achieve high classification accuracy on complex datasets, did not require much hyper-parameter tuning and has obtained good accuracy in analysing users’ attention. Extra Trees, similar to RF but with additional randomness introduced in the construction of DTs, making each tree more independent and random, showed good sensitivity and accuracy in the domain of situational memory.
- PIA23 (Piazzalunga et al., 2023) worked out a solution predicting the risk of subjects developing writing difficulties or learning disabilities. They applied CatBoost, a gradient boosting DT algorithm specifically designed to handle category-based features and widely used in various data mining and prediction tasks and compared it to an RF, a Gaussian Naïve Bayes (GNB), and a Support Vector Machine (SVM). CatBoost showed the best results and remained stable across validation and test data, even though overfitting due to the small dataset made the classifier perform poorly on the test data. This demonstrated the algorithm’s ability to make generalised predictions for unseen data. The authors showed that the algorithm could accurately

identify children who were at risk, especially when using game performance data as a predictor variable.

- The work presented in GIA19 (Giannakos et al., 2019) focused on using multimodal data to predict learning behaviour based on interaction with the gaming environment. Therefore, they chose an RF approach to predict the scores of each gaming session. In section 3.3.3, we will further explain the data explored.
- MAH22 (Mahboobeh et al., 2022) tested three ML classifiers in the environment of a platform for motor assessment of patients with Parkinson's Disease (PD), used to infer the stage of the disease. They compared RF, k-NN and SVM. Results showed that RF performed better when smaller feature vectors were applied, but was generally outperformed by the other classifiers (details below in sections b) and g)). Its performance range was between 81.5 % and 100 %, depending on the feature vector scenario.
- KAR21 (Karapapas & Goumopoulos, 2021) tested the performance of seven ML algorithms for the detection of mild cognitive impairment with a set of serious games: DT, RF, LR, SVM, k-NN, GNB, and MLP. RF performed well for manually selected features but worse for automatically selected features. DT was the second worst in both cases. We will indicate in the following sections the performances achieved with the rest.
- GOU21 (Goumopoulos et al., 2021): Six machine learning algorithms (DT, RF, SVM, LR, GNB, and k-NN) were compared for the classification of individuals as either cognitively impaired or healthy, based on performance data from serious games. The DT algorithm had high sensitivity but the lowest specificity, meaning it struggled to correctly classify healthy individuals. RF improved on DTs overall accuracy and specificity, but still did not match SVM, which showed the best overall performance (see details in section c)).
- NAI23 (Nair & Sakthivel, 2023) compared a DT, an SVM, Linear Discriminant Analysis (LDA) and NB to a Convolutional Neural Network-based deep learning model (deep CNN) proposed to identify the exercise completion status of hand and arm rehabilitation exercises. The DT was generally outperformed by the other classifiers, especially by the deep CNN, see details in section f).
- In JIA23 (Jiang et al., 2023), the RT was compared to a hierarchical fuzzy inference system to assess the motor function of the upper limbs of stroke patients based on the upper limbs' ability indexes as input data. It showed poorer performance, with lower accuracy and F1 scores (Taha, 2015) than the fuzzy system (see details in section e)).
- In HEN20 (Henderson et al., 2020), the authors evaluated five algorithms (RF, LR, SVM, GNB, and MLP) for recognising the affective states of students engaged with a game-based learning environment, using different multimodal data fusion techniques (details in section 3.3.3). RF was generally outperformed by the other methods for both unimodal and multimodal affect detection, being only occasionally the best-performing classification technique.

b) Rule-Based Systems

Fru17 (Frutos-Pascual & García Zapirain, 2017) mentioned 7 works that applied a Rule-Based System to user assessment; here, no recent works have been found.

c) Linear Classifiers

Compared to game control, there seem to be many more applications for linear classifiers in user assessment-related tasks. Regarding Logistic Regression (LR), the meta-review found only 4 works. In recent works, it was frequently used for comparison, but no work proposed this method as the best one.

- In KAR21 (Karapapas & Goumopoulos, 2021), GOU21 (Goumopoulos et al., 2021) and MEZ22 (Mezrar & Bendella, 2022), LR was applied for mild cognitive impairment detection. The outcome was that in manually selected datasets, sensitivity was excellent, but the performance was poor, while in automatically selected datasets, it was the other way around. This suggests that the performance of LR may vary across datasets and task contexts.
- In HEN20 (Henderson et al., 2020), LR was outperformed by MLP and SVM.

The application of Support Vector Machine (SVM) was much more frequently detected in the meta-review than LR, Fru17 (Frutos-Pascual & García Zapirain, 2017) cited 5 works, Kha21 (Khakpour & Colomo-Palacios, 2021) 4 works and Abd22 (Abd-alrazaq et al., 2022) 15 works.

In the scoping review, ten recent works have been found testing it for user assessment tasks: three of them focus on physical assessment (ESF22 (Esfahlani et al., 2022), JIA23 (Jiang et al., 2023), NAI23 (Nair & Sakthivel, 2023)); four on cognitive assessment (KAR21 (Karapapas & Goumopoulos, 2021), MAH22 (Mahboobeh et al., 2022), GOU21 (Goumopoulos et al., 2021), MEZ22 (Mezrar & Bendella, 2022)); and the remaining 3 on learning disabilities (PIA23 (Piazzalunga et al., 2023)), speech disorders (TOK23A (Toki et al., 2023a)) and affect detection (HEN20 (Henderson et al., 2020)).

The most important findings of those studies have been.

- ESF22 (Esfahlani et al., 2022): In this work, SVM and k-NN were trained with upper extremity joints' kinematic data and hand gestures to assess the subject's functional improvements through interacting with a video game for neurorehabilitation. Both showed the ability to recognise complex and noisy patterns in upper extremity data. While k-NN was more precise (see section g)), for SVM a smaller training set was sufficient.
- JIA23 (Jiang et al., 2023): SVM was used to compare with the proposed FL system and showed poorer performance with lower accuracy and F1 scores (Taha, 2015) (see section e) for details).
- NAI23 (Nair & Sakthivel, 2023): SVM performed better than DT and NB. A marginal performance improvement was observed when performing PCA-based dimensionality reduction as a pre-processing step or to identify the completion status of hand and arm rehabilitation exercises.
- KAR21 (Karapapas & Goumopoulos, 2021): The authors proposed a methodology based on implementing a support vector machine method that, along with a Gaussian NB (GNB) algorithm, outperformed the other five algorithms tested (mentioned in the former section). The algorithm could efficiently handle the mixed feature scope (in-game and subjective data) and showed endurance in the overfitting risk. On the other hand, GNB is a well-known classifier that is simple and can handle both discrete and continuous data, achieving a high performance even when the training dataset is limited. For simplicity, in Table 3 we count the GNB as a general NB.
- MAH22 (Mahboobeh et al., 2022): The SVM has a wider performance range than the classifiers it was compared to (k-NN and RF), depending on the scenario of feature vectors given, ranging from 88.9 up to 95.2 %.
- GOU21 (Goumopoulos et al., 2021): The SVM algorithm performed best, achieving an accuracy of 91.79 %, a sensitivity of 93.20 % and a specificity of 90 %. This indicates that SVM was the most effective at distinguishing between healthy individuals and those with mild cognitive impairment (MCI).
- PIA23 (Piazzalunga et al., 2023): SVM performed reasonably well during training but struggled with generalisation on test data compared to CatBoost and Random Forest, particularly with eye-tracking and drawing data, which indicates overfitting. Nevertheless, it performed slightly better than GNB.
- TOK23A (Toki et al., 2023a) is a study that aimed to detect neurodevelopmental disorders in children. Researchers tested various neural network optimisers such as Broyden-Fletcher-Goldfarb-Shannon (BFGS), Genetic Algorithm (GA), Particle Swarm

Optimisation (PSO), and Adaptive Moment Estimation (ADAM) on an ANN. The results were then compared with SVM, k-NN and an Integer-bounded Neural Network (INN). SVM and k-NN had poorer performance, with SVM achieving lower accuracy and suffering from generalisation issues compared to the INN (see details in section f)).
- In HEN20 (Henderson et al., 2020), the SVM was compared with RF, LR, GNB, and MLP. It performed well in both unimodal and multimodal settings; however, for the state of confusion, SVM's performance was lower compared to other algorithms, especially MLP (see section f)).

d) Nonlinear Classifiers

Almost every review included in the meta-review cited at least one work using Naïve Bayes (NB) or some variation. Fru17 (Frutos-Pascual & García Zapirain, 2017) found 11 works, but the more recent reviews found much less. Hidden Markov Model (HMM) was found less frequently than NB, but more in the earlier reviews than in the newer ones.

In the scoping review, the following works have been found that compared NB with other classifiers; no work proposed NB as the best-performing algorithm.

- In the arm rehabilitation recognition system presented in NAI23 (Nair & Sakthivel, 2023), the NB classifier used was not suitable for dealing with Histogram of Oriented Gradients (HOG) features extracted from motion waveform images but performed well in dealing with other features.
- In MEZ22 (Mezrar & Bendella, 2022), it was applied to the detection of early cognitive impairment and did not achieve the best results.
- KAR21 (Karapapas & Goumopoulos, 2021) tested a GNB algorithm in comparison with six other ML classifiers (see section a)) for the detection of mild cognitive impairment and found it performed similarly to the SVM and better than the rest.
- In GOU21 (Goumopoulos et al., 2021), GNB had the lowest accuracy and specificity compared to the other five classifiers tested, generating more incorrect classifications of both healthy and impaired individuals, though its sensitivity was comparable to SVM which had the overall best performance (see details in section b)).
- In HEN20 (Henderson et al., 2020), GNB had occasional strong performances but was generally outperformed by MLP and SVM.
- In PIA23 (Piazzalunga et al., 2023), GNB performed worst in comparison to RF, an SVM and the CatBoost algorithm, which performed best (see section a)).
- In LLO24 (Llorente et al., 2024), Bayesian Networks were used to detect deviations in cognitive function among patients with Alzheimer's disease.

e) Fuzzy Logic (FL)

FL was frequently cited in the reviews but less than for game control-related tasks, and more frequently in earlier years. In the scoping review, only one work was found working on a user-centred purpose.

- JIA23 (Jiang et al., 2023) designed membership functions and fuzzy rules for motor features in combination with the opinions of rehabilitation therapists to build a hierarchical fuzzy inference system to assess the motor function of upper limbs in stroke patients based on the upper limbs' ability indexes as input data. Results were compared to those of an SVM, a k-NN and an RT, outperforming all of them in accuracy, recall, precision and F1 score (Taha, 2015) for all tests.

f) Artificial Neural Networks (ANN)

Results of the meta-review show that ANN and variations (mostly CNN and Deep NN) were more frequently applied to user assessment than to game control. In the scoping review, four works have been found applying a shallow NN.

- ESF19 (Esfahlani et al., 2019) used the ANN algorithm to detect complex nonlinear correlations between data collected from the extremities to learn from the data and predict the improvement of the patient after using an SG for neurorehabilitation. The prediction accuracy achieved was 94 %.
- In KAR21 (Karapapas & Goumopoulos, 2021), the Multi-Layer Perceptron (MLP) was chosen as one of the test algorithms for the detection of mild cognitive impairment. MLP is a basic feed-forward ANN model that is commonly used to solve various types of problems, including classification and regression tasks. The experimental results showed that MLP exhibits extremely high sensitivity and specificity in manually selected datasets. However, in the automatically selected dataset, MLP showed lower sensitivity.
- In HEN20 (Henderson et al., 2020), the MLP was compared with SVM, RF, GNB and LR, and consistently performed well across most affective states, particularly when using multimodal data fusion techniques. For different settings, it was the most frequently optimal classifier, particularly in multimodal fusion, due to its ability to capture complex nonlinear relationships between data channels. Details of the multimodal data fusion techniques are given in section 3.3.3.
- In UPA23 (Upadhyay, 2023), the study introduced MemSpark, a system that integrates ANNs with VR for non-invasive treatment and assessment of dementia. The MemSpark analyser adopts an MLP architecture and demonstrates strong performance across multiple metrics, particularly in simulating ADAS-Cog standardized tests with high accuracy and regression adaptability. Compared to conventional approaches, the system more effectively slows cognitive decline and improves patients' quality of life.
- TOK23A (Toki et al., 2023a) applied different optimisers to an ANN (details in section b)) and concluded that the Genetic Algorithm (GA), which has a robust global search capability, was the most accurate one in the game score dataset. While Particle Swarm Optimisation (PSO) and Broyden-Fletcher-Goldfarb-Shannon (BFGS) produced low results in all instances, ADAM showed good classification accuracy and slightly outperformed the SVM and k-NN in the eye tracking dataset and the game score dataset. However, when screening for speech and verbal communication deficits in typically and atypically developing children, the Integer Bounded Neural Network (INN) achieved the best performance in comparison, attaining the optimal average classification error.

Four works applied a CNN, which is a CNN with convolutional layers, for processing and analysing data with a grid-like structure. They are mainly applied to computer vision tasks such as image recognition, image classification, target detection, semantic segmentation, etc.

- SVO20 (Svoren et al.) presented a multimodal dataset collected from different sensors (see details in section 3.3.3) to explore the possibility of creating emotionally aware ML algorithms that observe the players. Preliminary tests were presented with a CNN and part of the data set. The authors proposed its usage for the training of emotionally intelligent machines using reinforcement learning.
- NAS20 (Nasri et al., 2020) used a deep CNN in combination with gesture recognition models for high-accuracy gesture recognition by processing surface Electromyography signals, to improve the sensing of the user's gestures by the game used for rehabilitation. In the training process, the system achieved 99.8 % in training accuracy and 99.48 % in validation accuracy. The proposed network was tested with new subjects and obtained 82.15 % accuracy in gesture recognition.
- NAI23 (Nair & Sakthivel, 2023) proposed using motion time series as input to a CNN-based DL model to recognise the user's training completion status. The results were compared to DT, SVM, NB, and LD (see former sections) and indicate the superiority of the system as

indicated by the high accuracy (98.61 %), recall (0.99 %), precision (0.99 %) and F1-score (Taha, 2015) (0.99 %).

- In ABI24 (Abirami et al., 2024), the study proposed DeepMind Trainer—an innovative cognitive training system that utilises Convolutional Neural Networks (CNNs) to provide efficient rehabilitation support for individuals with cognitive impairments or neurodevelopmental disorders. Leveraging CNNs' powerful pattern recognition capabilities, the system interprets user input in real time and dynamically adjusts game difficulty, enabling personalised and adaptive cognitive training that enhances memory, attention, and creative thinking. The study highlights the critical role of CNNs in behaviour analysis, real-time feedback, and dynamic content generation, offering a novel technological pathway for cognitive rehabilitation.

A Deep Neural Network (DNN) is an NN with multiple hidden layers, and therefore 'deep', which enables it to learn more complex and abstract feature representations, thus improving the expressiveness and performance of the model.

- TOK23E (Toki et al., 2023b) investigated the efficacy of various machine learning approaches on biometric SmartSpeech datasets acquired during play to differentiate whether a child has a neurodevelopmental disorder, an autism spectrum disorder or attention deficit hyperactivity disorder. The authors compared the use of a Radial Basis Function (RBF) NN, a deep Neural Network (DNN), and a variation of Grammatical Evolution (GenClass). The results of this study showed that the best-performing classifiers for the eye tracking datasets were GenClass and DNN-4; for the heart-rate dataset, it was the RBF method, and for the game-based it were GenClass and RBF. As these are very specific methods, we classify them in Table 3 as a general DNN approach.
- In KIM21 (Kim, An, & Park, 2021), a DNN algorithm was developed to predict potential developmental disorders in children using drag-and-drop data and touch recordings as input. They successfully demonstrated the potential of a DNN to be used for screening developmental disabilities based on finger strokes used as biomarkers.

g) Kernel methods

One of the reviews (Abd22 (Abd-alrazaq et al., 2022)) cited 8 works applying k-NN to user assessment-related tasks. Here, six works have been found using k-NN, at least for comparison, e.g. for the prediction of multiple sclerosis (ESF22 (Esfahlani et al., 2022)), assessment of upper extremity motor function in stroke patients (JIA23 (Jiang et al., 2023)), determining the stage of neurodegenerative Parkinson's disease (MAH22 (Mahboobeh et al., 2022)) and detecting neurodevelopmental disorders in children (TOK23A (Toki et al., 2023a)), detecting cognitive impairment (GOU21 (Goumopoulos et al., 2021), KAR21 (Karapapas & Goumopoulos, 2021)). The most important findings of those studies have been.

- As already stated in section b), in ESF22 (Esfahlani et al., 2022), the k-NN showed higher accuracy than SVM, specifically 91.7 % vs. 88 %, and more robustness towards noisy training data.
- In JIA23 (Jiang et al., 2023), the k-NN approach was used for comparison with the proposed FL system and showed poorer performance with lower accuracy and F1 scores.
- In MAH22 (Mahboobeh et al., 2022), different scenarios of feature vectors generated from the fishing game were analysed using k-NN (and compared to SVM and RF). k-NN performed best across all scenarios, maintaining an accuracy rate higher than 90 %. It therefore demonstrated a very high stability compared to the rest.
- In TOK23A (Toki et al., 2023a), k-NN had a poorer performance than the other algorithms it was compared to (see section f)).
- In GOU21 (Goumopoulos et al., 2021), k-NN had a balanced performance with high sensitivity and specificity, almost matching SVM in its ability to classify both healthy and impaired individuals (90 %), though its overall accuracy was slightly lower (89.29 %).
- In KAR21 (Karapapas & Goumopoulos, 2021) k-NN was outperformed by NB and SVM, as noted in section b).

Only ESF22 (Esfahlani et al., 2022) and MAH22 (Mahboobeh et al., 2022) have been counted in Table 3 due to their good performance.

h) Unsupervised Learning

No works have been found that used any kind of unsupervised learning for user assessment.

i) Reinforcement Learning (RL)

RL, as an ML paradigm, focuses on learning by trial and error to achieve specific goals. The core idea is to maximise cumulative rewards by exploring the environment and adapting behaviours according to the reward signals obtained. As RL is very similar to the way humans learn, one study has been found that incorporates RL algorithms to detect mild cognitive impairment.

- In ALJ19 (Aljumaili et al., 2019), two ML algorithms were utilised in conjunction with an SG to detect Mild Cognitive Impairment (MCI) on mobile devices. First, an RL bot was trained to simulate different levels of cognitive decline, thereby simulating in this way different players. Then, CNN classifiers were trained on data from the bots at different levels of performance with the result that two different cognitive stages could be distinguished with 86.4 % accuracy.
- M23 (M & Surendran, 2023) developed a cognitive assistant based on an A3C deep reinforcement learning algorithm to support chess players with cognitive impairments, demonstrating its effectiveness in guiding users through decision-making by predicting potential threats and offering explanatory, personalised feedback, rather than simply optimizing for game victory.

j) Others

Tree-based Pipeline Optimisation Tool (TPOT) is an Automated ML tool designed to automate the construction and optimisation of machine learning pipelines. The goal of AutoML is to simplify the machine learning process, making it more accessible and easier to implement. TPOT uses genetic programming techniques to explore and optimise a range of machine learning models, hyperparameter configurations, and data pre-processing steps to identify the best solution.

In HOU23 (Hou et al.), TPOT was employed to generate different ML models for classifying cognitive impairment in older adults using Heart Rate Variability (HRV) measures and game performance data. Three different classification pipelines were created by classifying them based on three data sets: HRV only, game performance data only, and a combination of both. For the first set, the generated pipeline included Recursive Feature Elimination (RFE) for feature selection, a binarizer for pre-processing, and a Decision Tree Classifier, achieving an accuracy of 68.75 %. Using only game performance features, the pipeline used a Radial Basis Function (RBF) sampler and a Decision Tree Classifier and performed better with an accuracy of 83.33 %. The combination of both data sets did not improve accuracy (81.2 %); here, TPOT also created a DT classifier. We will add, therefore, a DT to Table 3, and omit TPOT as a method in the section "Others".

### *3.3.3. Review of studies handling multimodal data*

Multimodal data refers to data that involves multiple modes or types of information. Each mode represents a different way of sensing or capturing information, such as text, images, audio, video, or other types of signals. In other words, multimodal data integrates data from various sources or modalities to provide a more comprehensive representation of a phenomenon or an event.

Multimodal approaches in various research studies have significantly advanced our understanding of human behaviour, particularly in the realms of neurorehabilitation, gaming, education, and cognitive processes. These studies harness a variety of sensors and data sources to create a holistic understanding of the subjects under investigation. Therefore, we dedicate a separate chapter to the analysis of works that apply multimodal data to game control and user assessment.

13 studies were found in the established search range, 9 of which have been mentioned according to the AI used in the former chapters, namely ESF22 (Esfahlani et al., 2022), UPA23 (Upadhyay, 2023), LLO24 (Llorente et al., 2024), ABI24 (Abirami et al., 2024), ALS25 (Alsheikhy et al., 2025), DIA22 (Dias et al., 2022), DIA23 (Dias et al., 2023), GIA19 (Giannakos et al., 2019), and HEN20 (Henderson et al., 2020). In the following, we highlight the way they treated the multimodal data. All are focused on user assessment.

- ESF22 (Esfahlani et al., 2022) trained SVM and k-NN algorithms using the upper extremity (arm, forearm, and hand) joints' kinematic data obtained from Myo armbands and Microsoft Kinect sensors of participants while interacting with the developed video games.
- In UPA23 (Upadhyay, 2023), the MemSpark system employs multimodal isomorphic transfer learning, transferring knowledge from an audio-based cognitive assessment model to a VR game interaction model. Eight cognitive biomarkers (memory, reasoning, executive functions) and VR gameplay metrics (task time, accuracy) are aligned via Weighted Average Statistical Analysis (WASA) and Kernel PCA for integrated cognitive modelling.
- In LLO24 (Llorente et al., 2024), the PROCare4Life project integrates cognitive game data, wearable sensor data, and patient self-reports to model multidimensional cognitive states. A multimodal fusion engine generates a Cognitive State Score (CSS) and estimates cognitive decline probability (DCA) using a Bayesian network to support personalised intervention and remote health monitoring.
- ABI24 (Abirami et al., 2024) trains a convolutional neural network (CNN) on multimodal inputs, including camera images, touchscreen interactions, speech input, EEG, and heart rate signals. The model parses heterogeneous data in real time to dynamically adjust cognitive training game difficulty and optimise training outcomes.
- ALS25 (Alsheikhy et al., 2025) builds a high-dimensional multimodal dataset of motor, physiological, emotional, and VR system parameters to train an FFO-Bi-LSTM model for stroke rehabilitation assessment. Bi-LSTM captures temporal dependencies, while the FFO algorithm optimises hyperparameters and feature selection for precise, adaptive intervention.
- DIA22 (Dias et al., 2022) recorded Electrocardiography (ECG), Electromyography (EMG), respiration, temperature, electrodermal activity (EDA), Electroencephalography (EEG), Functional Near-Infrared Spectroscopy (fNIRS), and blood volume pulse (BVP), as well as movement data and decomposed them using Swarm Decomposition. The result trains a CNN to classify the physical and mental stress. This information is finally used to adapt the difficulty level of an SG to the user's affective state.
- DIA23 (Dias et al., 2023) is very similar, based on EDA, EEG, fNIRS, and BVP, which are decomposed by Swarm Decomposition and fed into a CNN. In this case, the user controls the serious neurogame (SNGs) EmoSense with their emotions.
- GIA19 (Giannakos et al., 2019) collected sensor data from five different sources: keystrokes (representing click-stream data), eye-tracking, EEG, video, and wristband (with sensors for heart rate, blood pressure, temperature and electrodermal activity levels) to predict learning behaviour. They explored how much the prediction could be improved compared with only using the number of keystrokes. They applied an RF algorithm for prediction and compared the results for different combinations of the multimodal data. The least prediction error (6 %) is obtained when combining eye-tracking, EEG and video, while cheaper settings as i.e. using webcam data plus keystrokes, already lead to an error rate of 15 %, which is much less than only using keystrokes (39 %).
- In the realm of affect detection in game-based learning environments, HEN20's study (Henderson et al., 2020) sought to find out which data channels are most predictive and how they should be combined. They employed spatial and temporal posture-based features obtained from a Kinect camera, together with gameplay interaction logs. After normalisation, features were utilised to train, validate, and test several ML-based models (SVM, RF, NB, LR, and MLP). Three variations of data fusion techniques were tested: two early fusions (combining features) and one late fusion (combining decisions of separately trained models). The MLP performed best for a majority of cases (60 %), and it robustly modelled complex, nonlinear relationships between modalities. The predictive value of each modality varies across affective states, so authors suggest that utilising dedicated models for each affective state, rather than inducing a single model to classify all affective states, could lead to improved detector performance.

The other 5 approaches have not appeared in the first three search queries because either they are not proposing a concrete AI architecture, or they are not focused on health or SG applications. Nevertheless, since their purpose aligns closely with our investigated RQs, we have included them in our analysis. In this part, all works also focus on user assessment.

- MIT22 (Mitsis et al., 2022) collected data from pressure sensors (mounted on a chair), heart rate monitoring, trajectory annotations, and in-game metrics, and performed statistical data analysis to identify the correlation between changes and the engagement as perceived by the player. Still, no AI has been employed to recognise engagement, but authors propose, as future work, to validate the multimodal feature vector through advanced feature fusion techniques via ML. They also propose to use this method on a constant feedback loop that enhances adherence to SG-based health interventions by maximising player engagement through generated game content.
- BOU19 (Boulton et al., 2019, pp. 3921–3927) collected different types of data, such as head posture, eye gaze, facial expression, game performance, and EEG while students played mobile games, providing insights into their attention and engagement. Machine learning was applied, but no specific algorithm was mentioned. The usage of Signal Detection Theory was crucial in segmenting the multimodal data into high and low-attention states. It enabled real-time assessment of engagement, providing accurate predictions of learner attention levels during game sessions. This algorithm was highly effective in real-time analysis and adaptive learning.
- SVO20 (Svoren et al.) collected data of different types from wristbands with four different sensors (obtaining electrodermal activity (EDA), inter-beat intervals (IBI), heart rate, blood volume pulse (BVP), motion information, and temperature), a webcam (obtaining the facial expression), and a game controller. Data is stored in a. csv file, no indications are given about data fusion. Preliminary tests are presented with a CNN and part of the dataset.
- MAT21 (Mat Sanusi et al., 2021) combined data from smartphones and motion sensors of a Kinect camera as input to an RNN to classify forehand table tennis strokes in a beginner's training environment. Data was stored in JSON format together with annotations and transformed into a tensor shape; the combination of different sampling frequencies was overcome by resampling. They explored whether the use of smartphones could be as accurate as the use of the Kinect camera and found that this is not the case, as the classification accuracy with Kinect data was better (64 % vs. 51 %).

Apart from the work presented by HEN20 (Henderson et al., 2020), who compared different data fusion techniques, all mentioned approaches combine the data features at an early stage, i.e. a fusion feature vector is created to be fed into the AI (early fusion). Another possibility is to train different models and to fuse at the end the decisions weighted by the reliability of each model (late fusion). This is what one work proposed:

SHA22 (Sharma et al., 2022) investigated the relationship between student interactions and learning experiences. The study collected data from eye-tracking, wristbands (with sensors for HRV, blood pressure, temperature, and EDA levels), and Kinect skeleton data. After normalisation, the data was analysed, and features were computed. For predictive modelling, the decisions of 7 different algorithms were combined as a weighted average (SVM with linear, radial, and polynomial kernels; Gaussian process models with linear, radial, and polynomial kernels; and M5 model trees). As this method does not specify a particular algorithm, it cannot be reflected in Table 3.

## 4. Discussion

The article by Dam23 (Damaševičius et al., 2023) is the most recent review of serious games (SGs) in healthcare, published approximately a year before the inception of this review. As a meta-review, it offers a comprehensive overview of the field. Dam23 (Damaševičius et al., 2023) noted limited research on sustaining player interest in healthy behaviours and motivation, which we will compare with our broader examination of games and gamification in general (not restricted to the health sector). They also observed a growing use of AI to personalise and adapt SGs and gamification interventions. The authors highlighted the need for further research to enhance user engagement and tailor games to diverse patient needs. These findings underscore the relevance of our review, which seeks to provide detailed insights into existing methods and potential areas for further development.

The response to RQ1 is illustrated in Table 3 and Fig. 4. Certain algorithms are notably more prevalent than others, with clear distinctions based on their intended purpose. Globally, the most frequently applied are in this order: Artificial Neural Networks (ANNs), Decision Trees (DTs), Fuzzy Logic (FL), Support Vector Machines (SVMs), and Naive Bayes (NB). However, the purpose is important: results show significant differences in frequency of usage, depending on whether the aim is to control the game difficulty or content or to assess the state of the user. While ANNs, DTs and FL are used for both purposes in nearly the same amount, SVM and NB are much more present in user assessment than in game control. Here, SVM wins with 90 % usage.

Comparing only the algorithms applied for game control, FL is the most commonly used method, both in earlier years and in recent years. DTs, ANNs, and Finite State Machines (FSM) were also frequently used, but the scoping review did not find any recent applications of them. However, Reinforcement Learning (RL), NB, and rule-based systems, which were less prevalent than the aforementioned algorithms, are still in use. Additionally, new techniques have emerged: three studies were found using CNNs ((Izountar et al., 2021), (Dias et al., 2022), (Dias et al., 2023)), and three studies based on k-means (Bouatrous et al., 2023), genetic algorithms (Kalafatis et al. et al., 2023), Machine Theory of Minds (MTOM) (Kempitiya et al., 2022) and GANs (Chen et al., 2024).

Regarding user assessment, the clear frontrunner is SVM. In the meta-review, it is the most frequently applied algorithm, and in the scoping review, we found nine studies using it for comparison. Of those, two studies found it performed best ((Goumopoulos et al., 2021), (Karapapas & Goumopoulos, 2021)), and another two found it to be the second best ((Mahboobeh et al., 2022), (Henderson et al., 2020)). Concerning neural networks, the meta-review found frequent use of general ANNs and occasional use of CNNs. This scenario appears to have evolved towards more specific applications, such as proposals to apply optimisers like ADAM or INN (Toki et al., 2023a), Toki et al., 2023bnd the usage of CNNs ((Svoren et al.)), deep CNNs ((Nair & Sakthivel, 2023), (Nasri et al., 2020)), MLP ((Kempitiya et al., 2022), (Karapapas & Goumopoulos, 2021), (Upadhyay, 2023), (Henderson et al., 2020)), Deep Dense Networks (DDN) ((Kim, An, & Park, 2021), (Toki et al., 2023b)) and RNNs ((Alsheikhy et al., 2025), (Mat Sanusi et al., 2021)). DTs ((Mezrar & Bendella, 2022), (Goumopoulos et al., 2021), (Hou et al.), (Karapapas & Goumopoulos, 2021), (Nair & Sakthivel, 2023)), RF ((Mezrar & Bendella, 2022), (Aljumaili et al., 2019), (Karapapas & Goumopoulos, 2021), (Mahboobeh et al., 2022), (Piazzalunga et al., 2023), (Giannakos et al., 2019), (Henderson et al., 2020)), NB ((Mezrar & Bendella, 2022), (Goumopoulos et al., 2021), (Karapapas & Goumopoulos, 2021), (Nair & Sakthivel, 2023), (Piazzalunga et al., 2023), (Llorente et al., 2024), (Henderson et al., 2020)), and k-NN ((Goumopoulos et al., 2021), (Mahboobeh et al., 2022), (Toki et al., 2023a), (Esfahlani et al., 2022), (Jiang et al., 2023)) are still in use, at least for comparisons, though they do not always achieve the best performance. Hidden Markov Models (HMM) and rule-based systems were no longer found in recent research. Two new approaches have emerged: Hyperseed ((Kempitiya et al., 2022)) and Catboost ((Piazzalunga et al., 2023)). It is worth noting that M23 (M & Surendran, 2023) presents a cognitive assistant based on the A3C deep reinforcement learning algorithm to support chess players with cognitive impairments. Uniquely, it employs a virtual agent to guide users through training by predicting threats and providing explanatory, personalised feedback. The focus is not on winning, but on enhancing decision-making and cognitive support.

Nevertheless, in the newer studies targeting user assessment, the classical ML techniques have been frequently used for comparison. We found 10 studies that compared 3 and 8 algorithms. Most frequently used in this order were: SVM, RF, k-NN, NB, LR and DT.

### *4.1. Analysis of algorithms used for game control (answering RQ2)*

Game control relies on a range of decisions based on the type and quality of the input data, which is obtained from the user to personalise their experience and meet their therapeutic needs. The earlier reviews (Fru17 (Frutos-Pascual & García Zapirain, 2017) and Zoh18 (Zohaib, 2018)) found contradictory results. While Fru17 (Frutos-Pascual & García Zapirain, 2017) found that DT, FSM, and FL were more commonly used, followed by ANN and HMM, Zoh18 (Zohaib, 2018), whose review was more specific for Dynamic Difficulty Adaptation (DDA), identified ANN as the most cited algorithm, followed by NB.

Later, Kha21 (Khakpour & Colomo-Palacios, 2021), similarly, observed different trends, focusing on the intersection of gamification and ML, though not specifically on SGs or health. Advanced methods like CNN, RNN, CN2 (rule-based), and ε-greedy (RL) were mentioned for game control.

In 2022, four selected reviews were published (Abd22 (Abd-alrazaq et al., 2022), Lu22 (Lu & Li, 2022), Saj22 (Sajjadi et al., 2022), and Lop22 (Lopes & Lopes, 2022)) but only Abd22 (Abd-alrazaq et al., 2022) and Lop22 (Lopes & Lopes, 2022) provided relevant data for our meta-review. Lop22 (Lopes & Lopes, 2022) focused on DDA and observed a preference for RL and DL, explaining that RL's ability to adapt in real-time without prior training makes it more flexible for DDA.

In 2023, four reviews were published (Sri23 (Srimathi & Anitha, 2023), Ayd23 (Aydin et al., 2023), Tou23 (Toukiloglou & Xinogalos, 2023), and Par23 (Paraschos & Koulouriotis, 2023)). Sri23 (Srimathi & Anitha, 2023) listed nine AI algorithms for processing gamification elements: LR, NB, SVM, RF, ANN, FL, ANFIS, DTs, and RL. Both Ayd23 (Aydin et al., 2023) and Tou23 (Toukiloglou & Xinogalos, 2023) examined the adaptability of educational games, with Ayd23 (Aydin et al., 2023) covering 26 articles (eight more than Tou23 (Toukiloglou & Xinogalos, 2023)). Ayd23 (Aydin et al., 2023) found rule-based systems most commonly used for game adaptation, while Tou23 (Toukiloglou & Xinogalos, 2023) identified NB as more frequent. Interestingly, Ayd23 (Aydin et al., 2023) also mentioned FL, DDN, Information Field Theory, and Item Response Theory, but found no use of DT or FSM.

Par23 (Paraschos & Koulouriotis, 2023) is the most relevant article in our meta-review, as it is the newest and most comprehensive, with 109 articles published over 17 years (2005–2021). Table 3 shows a wide range of algorithms used during this time. Unfortunately, the number of works is not as high as expected, as older reviews are not fully included. Its focus on adaptation and experience personalisation limits overlaps with other reviews. Few publications appeared in the first 10 years, with most techniques emerging after 2015, and only 24 articles mentioning AI. The variety of techniques shows no clear focus, contradicting our findings on game control and user assessment methods. The authors of Par23 (Paraschos & Koulouriotis, 2023) graphically illustrate AI's evolution from 2014 to 2018, revealing its early use to adapt to player behaviour, eventually leading to deep user assessment (e.g. affective state, interaction, physiological data) for real personalisation and motivation. After 2018, they anticipate significant advances in real-time content tailoring to boost engagement and motivation.

Further on, we will discuss the findings of the scoping review.

In AI designed for game control, most applications aim to leverage AI's ability to analyse players' game behaviour, decision-making patterns and skill choices to obtain game-relevant data metrics and automate game adjustments. By analysing data on the player's game performance and/or in-game behaviour, as well as other external data, AI can detect potential in-game problems and challenges, thus providing knowledge about different game strategies. This can be used to advise players to adjust their gaming strategies or adopt specific gaming behaviours to improve the efficiency of their rehabilitation, as well as to automatically improve the content and difficulty of the game to provide a better gaming experience and a higher level of autonomy. Personalised rehabilitation plans require a complex decision-making process.

In our research, we have found that in all studies targeting game control-centred approaches, whether it is the analysis of the user's game performance (EUN23 (Eun et al., 2023), STR22 (Streicher & Aydinbas, 2022), ESF19 (Esfahlani et al., 2019), KEM22 (Kempitiya et al., 2022), BOU23 (Bouatrous et al., 2023), KOS23 (Kostkova et al., 2023), KAL23 (Kalafatis et al. et al., 2023), GHO22 (Ghorbani et al., 2023), CHE24 (Chen et al., 2024), PEL24 (Pelosi et al., 2024)) or the analysis of the user's emotions (DIA22 (Dias et al., 2022), DIA23 (Dias et al., 2023), IZO21 (Izountar et al., 2021)), the goal is mostly to achieve appropriate difficulty level recommendations to improve the user's gaming experience and rehabilitation effectiveness. One exception is MAS23 (Maskeliunas et al., 2022): here, a DRL algorithm was applied to control the game flow and content by choosing scenarios and gaming events from a fixed set for each interaction. In our opinion, this kind of game control could open up many more possibilities to achieve real personalised therapies and higher motivation; nevertheless, the work of MAS23 (Maskeliunas et al., 2022) was created for caregivers, so we did not find any proposal that applied AI to changes of the environment of a therapeutic game.

For user sentiment analysis, CNNs are often used as the basis of the system (DIA22 (Dias et al., 2022), DIA23 (Dias et al., 2023), IZO21 (Izountar et al., 2021)), due to the strong feature extraction capability of CNNs, which can automatically extract higher-order features through data stacking, and their flexible structure, which can be adapted to different types and sizes of sentiment analysis tasks.

For analysing user game performance, rule-based learning (EUN23 (Eun et al., 2023)), FL (ESF19 (Esfahlani et al., 2019), GHO22 (Ghorbani et al., 2023)), NB (STR22 (Streicher & Aydinbas, 2022)), MLP (KEM22 (Kempitiya et al., 2022)), k-Means (BOU23 (Bouatrous et al., 2023)), RL (KOS23 (Kostkova et al., 2023), MAS23 (Maskeliunas et al., 2022), PEL24 (Pelosi et al., 2024)), genetic algorithms (KAL23 (Kalafatis et al. et al., 2023)) and CHE24 (Chen et al., 2024) have been applied. Researchers have chosen different algorithms based on different data and needs, with the emphasis that the algorithms need to be able to help the player maintain a state of mind flow, i.e., be fully focused on the game and strike an optimal balance between the challenges of the game and the player's abilities. This balance helps the player stick to the treatment program and stay motivated over a longer time. In this case, the player does not win all the games easily, so the game does not become too easy and boring, but gains enough motivation to continue playing. All of the data demonstrated the importance of balancing the challenge of the game. KEM22 (Kempitiya et al., 2022) applies a machine theory of mind (MTOM) framework to predict player mental states, aiming to simulate human cognition. Combining insights from AI, cognitive science, and psychology, MTOM holds promise for advancing intelligent and human-centric AI systems. CHE24 (Chen et al., 2024) explores both Generative Adversarial Networks (GANs) and Long Short-Term Memory (LSTM) models to generate personalised difficulty levels for stroke rehabilitation games. The GAN model uses user data to simulate diverse gameplay trajectories, while the LSTM model leverages time-series performance data to predict future difficulty. This demonstrates a novel and scalable method for intelligent game adaptation.

However, we also found that there are almost no AI applications with virtual players or opponents for serious games, but this could be a very interesting way to set up immersive game mechanics.

### *4.2. Analysis of algorithms used for user assessment and with multimodal data (answering RQ3)*

Fru17 (Frutos-Pascual & García Zapirain, 2017) found that user assessment primarily used NB, followed by DT, Rule-Based Systems, and FL/ANN equally. Surprisingly, the next significant review, Abd22 (Abd-alrazaq et al., 2022), came five years later, overlapping Fru17 (Frutos-Pascual & García Zapirain, 2017) for 2010–2014 (20 %) and extending it to 2021 (80 %). However, we could not extract enough data from Abd22 (Abd-alrazaq et al., 2022) to compare its findings with Fru17 (Frutos-Pascual & García Zapirain, 2017), leading us to re-review the cited works. Our analysis showed AI algorithms were mainly used for user evaluation, with SVM and ANN being most common (6 of 15 using CNN), likely due to a focus on healthcare and rehabilitation. Personalisation and adaptation were only briefly mentioned, despite their importance according to other authors.

Kha21 (Khakpour & Colomo-Palacios, 2021) similarly found more works on user evaluation than game control, with SVM commonly used to predict user performance and affective states. Other methods included kernel optimisation, clustering, and LR, mainly for personalizing learning applications by a) adapting tasks based on user affective states, b) predicting future behaviour to provide motivational feedback, and c) clustering users by capabilities to offer tailored gamified tasks. Tou23 (Toukiloglou & Xinogalos, 2023) also found FL prevalent for user assessment.

In our scoping review, we analysed SGs using AI and found that adaptive SGs for education, visualisation, and simulation training exhibit greater adaptability than those for rehabilitation. These games also use a wider range of AI algorithms to improve user experiences and learning outcomes.

In user-centred AI applications, AI primarily analyses player movement patterns, reaction times, and strategy choices for rehabilitation metrics. By assessing performance, physiological data, in-game behaviours (e.g., movement deterioration), and external data, AI can detect potential health issues and disease risks. This technology can advise players on preventative measures or medical help and assist therapists in diagnosing or predicting changes in disease progression.

Seven studies were identified using a Decision Tree (DT) or a variant for user assessment. In NAI23 (Nair & Sakthivel, 2023), a DT was used to model the system as a clear tree structure, visualising key decision paths and player behaviour, while also highlighting the importance of each feature. MEZ22 (Mezrar & Bendella, 2022) improved accuracy using the RF algorithm, which averages multiple DTs. RF trains each tree on random samples and features, with nodes split using a random subset of features, enhancing prediction accuracy. Extra Tree also performed well due to the high dimensionality and size of the dataset. In PIA23 (Piazzalunga et al., 2023), CatBoost proved effective for small datasets by preventing overfitting and generalising well to unseen data.

We found that SVM is the most frequently used algorithm (GOU21 (Goumopoulos et al., 2021), KAR21 (Karapapas & Goumopoulos, 2021), MAH22 (Mahboobeh et al., 2022), NAI23 (Nair & Sakthivel, 2023), PIA23 (Piazzalunga et al., 2023), TOK23A (Toki et al., 2023a) ESF22 (Esfahlani et al., 2022), HEN20 (Henderson et al., 2020), JIA23 (Jiang et al., 2023)), though it was not always the final choice. This is often due to the small size of experimentally generated datasets, which can lead to poor training results for SVMs. In such cases, SVM struggles to achieve high prediction accuracy (JIA23 (Jiang et al., 2023)) as smaller datasets may not provide enough data to effectively define the optimal hyperplane for classification. While SVM excels in large, well-structured datasets, particularly in tasks like image classification and text categorisation (MAH22 (Mahboobeh et al., 2022)), it requires large random samples to create a robust model. However, with multiclass SVM (NAI23 (Nair & Sakthivel, 2023)) or proper pre-processing (GOU21 (Goumopoulos et al., 2021)), even a small training set can be effective, provided the right parameters are selected for testing accuracy rather than relying on traditional models with large datasets.

LR and SVM are both linear models, but LR cannot handle non-linearly differentiable data directly. KAR21 (Karapapas & Goumopoulos, 2021) and MEZ22 (Mezrar & Bendella, 2022) show that data pre-processing is required when using LR.

We found eight articles dealing with different ANN models. They excel at handling complex non-linear relationships, making them suitable for serious games. CNN, as DL models, perform well with spatial data, efficiently capturing features through their hierarchical structure (NAS20 (Nasri et al., 2020), NAI23 (Nair & Sakthivel, 2023)). MLP models usually require input data standardisation or normalisation. HEN20 (Henderson et al., 2020) and KAR21 (Karapapas & Goumopoulos, 2021) indicate that MLPs, due to their many parameters, are prone to overfitting on small datasets and need tuning to improve the generalisation of new data. DNNs can learn complex features from high-dimensional datasets like physiological metrics and movement patterns, aiding in rehabilitation assessment, but require extensive data, time, and computational resources (NAS20 (Nasri et al., 2020), KIM21 (Kim, An, & Park, 2021), TOK23A (Toki et al., 2023a), TOK23E (Toki et al., 2023b)).

K-NN was used for classification and regression in six studies. Its simplicity makes it popular, but its performance depends on the k-value and distance metric used, varying by dataset. In studies like MAH22 (Mahboobeh et al., 2022) and ESF22 (Esfahlani et al., 2022), k-NN performed well, likely due to its ability to capture patterns in those datasets. However, in studies like TOK23 (Toki et al., 2023a), its performance was more general, possibly due to dataset complexity. K-NN works best with small, low-dimensional datasets but may require pre-processing or feature engineering to improve performance.

RL, although frequently found in previous research, has been found now in only one study: ALJ19 (Aljumaili et al., 2019) extended their case dataset using RL exploring its ability to simulate complex systems. This algorithm can be extremely helpful for training in case of insufficient samples.

In our research, we have found that in all studies targeting game control-centred approaches, whether it is the analysis of the user's game performance (EUN23 (Eun et al., 2023), STR22 (Streicher & Aydinbas, 2022), ESF19 (Esfahlani et al., 2019), KEM22 (Kempitiya et al., 2022), BOU23 (Bouatrous et al., 2023), KOS23 (Kostkova et al., 2023), KAL23 (Kalafatis et al. et al., 2023)) or the analysis of the user's emotions (DIA22 (Dias et al., 2022), DIA23 (Dias et al., 2023), IZO21 (Izountar et al., 2021)), the goal is mostly to achieve appropriate difficulty level recommendations to improve the user's gaming experience and rehabilitation effectiveness. All the data demonstrated the importance of balancing the challenge of the game. Data mining techniques, such as cluster analysis and association rule mining, can extract patterns from large datasets to inform personalised rehabilitation plans. Reinforcement learning algorithms adjust rehabilitation plans based on player performance and feedback to optimise outcomes. Optimisation algorithms, such as simulated annealing and linear programming, determine the most effective rehabilitation plan, while genetic algorithms continually evolve and optimise these plans. Although some algorithms are not commonly used in recent studies, this does not mean they are outdated, as they may still be relevant depending on the dataset.

We also found significantly fewer studies in 2020 and 2021 compared to other years, likely due to the impact of the COVID-19 pandemic, which made it difficult to conduct experiments.

When selecting an AI algorithm, trade-offs must be made based on the problem's specific characteristics. Algorithms may vary in performance depending on the dataset and problem. Factors such as applicability, dataset size, feature complexity, computational demands, and result requirements should guide the choice of the most suitable algorithm. Additionally, algorithm characteristics like interpretability, stability, and scalability should be considered to ensure reliability and effectiveness in practical applications.

#### *4.2.1. Usage of multimodal data sets*

In our analysis, we paid special attention to the application of multimodal data, as this seems to be a promising approach. The type of data used is manifold, as there are body movement data (ESF22 (Esfahlani et al., 2022), DIA22 (Dias et al., 2022), LLO24 (Llorente et al., 2024), SVO20 (Svoren et al.), SHA22 (Sharma et al., 2022) and MAT21 (Mat Sanusi et al., 2021)), physiological signals including electrocardiogram ECG, electromyogram EMG, skin electrical activity EDA, etc. (DIA22 (Dias et al., 2022), DIA23 (Dias et al., 2023), LLO24 (Llorente et al., 2024), GIA19 (Giannakos et al., 2019) and SVO20 (Svoren et al.)), EEG data (DIA22 (Dias et al., 2022), DIA23 (Dias et al., 2023), GIA19 (Giannakos et al., 2019) and BOU19 (Boulton et al., 2019, pp. 3921–3927)), eye-tracking data (SHA22 (Sharma et al., 2022), GIA19 (Giannakos et al., 2019)), facial expression data (GIA19 (Giannakos et al., 2019), BOU19 (Boulton et al., 2019, pp. 3921–3927), SVO20 (Svoren et al.)), heart rate monitoring data (MIT22 (Mitsis et al., 2022), LLO24 (Llorente et al., 2024), ABI24 (Abirami et al., 2024), SVO20 (Svoren et al.), SHA22 (Sharma et al., 2022)), temperature and blood volume pulse (BVP) data (DIA22 (Dias et al., 2022), DIA23 (Dias et al., 2023), GIA19 (Giannakos et al., 2019), SVO20 (Svoren et al.), SHA22 (Sharma et al., 2022)), HEN20 (Henderson et al., 2020), LLO24 (Llorente et al., 2024), ABI24 (Abirami et al., 2024) and UPA23 (Upadhyay, 2023) used game interaction logs, and MIT22 (Mitsis et al., 2022) used trajectory annotation data.

The use of two data fusion techniques, early fusion and late fusion, is mentioned in some studies involving multimodal approaches. Early fusion refers to the fusion of features from different modalities to create a unified feature vector, which is later input into the AI model for training. ESF22 (Esfahlani et al., 2022), DIA22 (Dias et al., 2022), DIA23 (Dias et al., 2023), UPA23 (Upadhyay, 2023) and GIA19 (Giannakos et al., 2019) used early fusion. And late fusion is the process of training separate models for each modality's data and then fusing the outputs of each model at the end; SHA22 (Sharma et al., 2022) used late fusion, where the decisions of seven different models were weighted and averaged for fusion, and LLO24 (Llorente et al., 2024) and HEN20 (Henderson et al., 2020) partially used early fusion and partially used late fusion. Among the reviewed studies, intermediate fusion was employed in only one case ALS25 (Alsheikhy et al., 2025).

Strong inter-modal correlations and interactions need to be considered when choosing a data fusion technique. In ESF22 (Esfahlani et al., 2022), both Myo armband and Kinect sensors were used to capture upper limb motion data. The data captured by the different sensors all reflect arm, forearm, and hand movements, and thus these data are somehow complementary and strongly correlated with each other. Early fusion can integrate these correlations at the feature level and help the model to better understand the overall movement pattern. Physiological signals (e.g., ECG, EEG, EDA, etc.) are used in both DIA22 (Dias et al., 2022) and DIA23 (Dias et al., 2023), and there are complex interactions between these signals, e.g., emotion and stress can affect both heart rate and electrical skin activity. Early fusion allows these interactions to be handled directly in the model, helping to recognise the user's emotional state. The temporal consistency of the data is likewise a major key in choosing a data fusion technique; eye tracking, EEG, facial expression and physiological data (e.g., heart rate, electrical skin activity, etc.) in GIA19 (Giannakos et al., 2019) are all captured at the same time step. In this case, early fusion can integrate these synchronised data to form a unified feature representation without losing temporal information. Moreover, the study of GIA19 (Giannakos et al., 2019) combines eye tracking, EEG and physiological data, which are of high dimensionality, and unified processing can simplify the subsequent training steps.

In contrast, in SHA22 (Sharma et al., 2022), the physical meaning and data characteristics of data such as eye tracking, physiological data from wristbands (e.g., heart rate variability, blood pressure, body temperature, and electrical skin activity), and Kinect bone data are more different, and later fusion can allow for independent processing and optimisation of a certain modality to avoid confusing or interfering information. Moreover, these data may have been acquired at different frequencies at the time of acquisition, and post-fusion allows for separate processing of these asynchronous data, avoiding loss of features or incorrect combinations due to timing inconsistencies. Moreover, post-fusion allows the construction of separate models for each modality of data, and by processing them independently, SHA22 (Sharma et al., 2022) optimises for each modality, improving the model's ability to handle all types of data. SHA22 (Sharma et al., 2022) uses several different algorithms, including SVMs with different kernel functions, Gaussian Process Models, and M5 model trees. Posterior fusion allows the predictions from these different models to be integrated, taking full advantage of the strengths of each algorithm. Each algorithm may perform better when dealing with specific types of data, and through post-fusion, the strengths of multiple algorithms can be combined to enhance the combined performance of the models.

An increasing number of researchers are incorporating multimodal data into their studies to comprehensively assess and analyse complex phenomena. Multimodal approaches integrate information from various data sources, such as visual, auditory, physiological signals, and behavioural data, enabling the capture of richer features and dependencies, thus enhancing model performance and predictive accuracy. This cross-disciplinary approach not only addresses the limitations of single-modal data but also provides more precise and personalised insights.

Overall multimodal data provides the possibility of explaining the same phenomenon from multiple perspectives because of the collection of data from different modalities, and when one modal data is unreliable, the data from other modalities can be supplemented, which can be a good way to improve the stability of the overall system, and further enhance the performance of the model in various types of tasks. However, it is also due to the large differences in the data forms, sampling rates, time steps, etc. of different modalities, the effective integration of these heterogeneous data becomes a difficult point, and the high-dimensional features also increase the computational complexity, and because of the complexity of the multimodal data, the selection of appropriate fusion strategies also becomes a key point.

### 4.3. Evolution and future trends

With the development of AI technology and the popularisation of electronic devices, more and more researchers have begun to subjectively integrate AI technology with other disciplines. Among them, the combination of AI and serious gaming has shown great potential in the field of building wellness, providing innovative ways for personalised interventions, state assessment, and long-term rehabilitation management. In this context, research has developed a set of open-source AI components covering emotion recognition, difficulty adaptation, natural language processing and trusted NPC modelling (Westera et al., 2020). This system has been validated in serious health games and has cross-platform reusability, which can significantly reduce the development cost and cycle time.

Key areas of innovation include **narrative-driven games**, **immersive virtual environments**, and **biofeedback-based mechanics**. While game flow control and adaptive gameplay have been well explored in entertainment games (Aan et al., 2024; Chodvadiya et al., 2024; Winther et al., 2023; Yong et al., 2023), their application to therapeutic contexts remains limited. Designing emotionally engaging and personalised interactive experiences may enhance user motivation, adherence, and the overall efficacy of rehabilitation programs.

The rise of **Generative AI (GenAI)** introduces transformative possibilities for SGs. These models can dynamically generate storylines, training scenarios, visual environments, or dialogue content based on individual profiles (Summerville et al., 2017). By combining real-time physiological and behavioural data, GenAI systems can tailor the game's challenge level, narrative elements, and pacing, offering true *on-the-fly customisation* that aligns with a patient's moment-to-moment status and therapeutic needs ((Volz et al., 2018, pp. 221–228), (Miller et al., 2020)).

Future SG systems will increasingly rely on **multimodal data fusion**, supported by advances in **wearable technologies** and **extended reality (XR)** (VR/AR) (Nasri, 2025b). Wearables enable continuous monitoring of vital signals (e.g., heart rate, gait, electrodermal activity), while XR can simulate realistic environments to promote engagement and real-world transfer of training. Multimodal AI models will be essential to integrate voice, motion, facial expressions, and contextual interactions into unified representations of user state (Martin et al., 2021).

However, the fusion of these technologies also brings significant challenges. **Data heterogeneity**, **temporal complexity**, and **privacy concerns** outpace the capabilities of traditional unimodal or static AI models. Consequently, future systems must adopt **multimodal deep learning architectures**, **federated learning** to preserve privacy across decentralised data sources, and **adaptive reinforcement learning** to personalise interventions over time. These advanced methods are needed not only to improve accuracy but also to ensure clinical **relevance**, **robustness**, and **trustworthiness**.

In this context, **interdisciplinary collaboration** will be a key consideration in future research, as the interdisciplinary nature of rehabilitation games requires collaboration between researchers, game developers, therapists, and healthcare professionals. A holistic approach that incorporates insights from various fields is critical to the development of effective rehabilitation interventions.

Finally, as SGs become more clinically integrated, the need for **transparent and interpretable AI systems** becomes increasingly urgent. Understanding how and why algorithms make decisions is critical for clinical accountability. Techniques from **Explainable AI (XAI)** will be essential to clarify algorithmic reasoning and build trust among users and practitioners. In parallel, safeguarding personal health data, particularly as GenAI and real-time analytics are deployed, must remain a top ethical priority ((Martin et al., 2021), (Berger & Muller, 2021), (Clement et al., 2023)).

### 4.4. Limitations

We acknowledge several limitations in our review methodology that may affect the comprehensiveness and accuracy of our findings.

For the meta-review, synthesising content from previously published reviews introduces inherent risks of bias. The selected reviews varied considerably in scope, narrative focus, terminology, and publication timelines. Some reviews presented ambiguous or incomplete descriptions, which may have led to misinterpretations or inconsistencies in classification. Moreover, the original authors may have omitted important methodological details or misclassified the studies they reviewed. To address these issues, we consulted many of the primary studies referenced in the reviews, particularly when algorithmic methods were not explicitly stated. Despite this, identifying and categorising AI techniques remained difficult. Many reviews referred only to broad categories such as “machine learning” or “artificial neural networks,” without specifying algorithms. In some cases, feasibility studies lacked technical details, or multiple models were compared without a clear indication of the best-performing approach. Certain studies cited proprietary software or models that could not be matched to known AI methods and were therefore excluded. Additionally, some articles may have used AI techniques without explicitly labelling them as such, leading to possible omissions. Challenges also arose in classifying algorithms as either user assessment or game control, especially when adaptations were based on simple performance metrics versus deeper user modelling. Lastly, there may have been overlapping primary studies cited across multiple reviews, which we were not always able to detect.

For the scoping review, additional limitations apply. Studies that referenced AI methods without specifying their purpose or the algorithms used were excluded, potentially omitting relevant contributions. Our search was limited to selected academic databases, and as a result, relevant studies in less widely indexed journals may have been missed. Furthermore, we did not include grey literature, preprints, or non-peer-reviewed sources, which may be particularly relevant in the rapidly evolving domain of AI applications. Language restrictions may also have excluded valuable studies published in non-English languages. Finally, as with any scoping review, the results are dependent on our predefined inclusion criteria and search strategy. Despite efforts to design a comprehensive query, it is possible that studies using AI in unconventional or interdisciplinary ways were inadvertently excluded.

## 5. Conclusions

This enhanced meta-review provides a comprehensive analysis of how artificial intelligence (AI) is applied in serious games (SGs) for healthcare, focusing on two main areas: game control and user assessment. In doing so, we aimed to outline a “landscape” of AI usage in each domain.

In the area of **user assessment**, AI serves two primary functions: (1) gathering basic performance data to adapt gameplay in real time, and (2) conducting in-depth analyses of the user’s health or emotional state to support medical evaluation and personalised interventions. The first function typically involves measuring a fitness function and can be implemented during gameplay. The second requires physiological or emotional data captured through sensors, with machine learning (ML) methods—especially Support Vector Machines (SVMs)—commonly employed. However, the accuracy of these methods depends heavily on the dataset size and quality. Artificial Neural Networks (ANNs), often enhanced with optimisers like ADAM and models such as Integer Bounded Neural Networks (INNs), are also increasingly used, particularly for estimating performance and emotional states. Notably, most disease prediction models focus on player behaviour, indicating an opportunity for broader healthcare applications.

In the domain of **game control**, AI algorithms have evolved to support more innovative and personalised gameplay. While Fuzzy Logic (FL) remains consistently used, traditional methods like Decision Trees (DTs), Reinforcement Learning (RL), ANNs, Naïve Bayes (NB), and Finite State Machines (FSM) have declined in popularity. In contrast, emerging techniques—including Convolutional Neural Networks (CNNs), k-means clustering, Genetic Algorithms (GAs), and Machine Theory of Mind (MTOM)—are gaining momentum. These are primarily used to adapt difficulty levels, though only one study has explored simultaneous adaptation of both content and gameplay. The advent of generative AI appears especially promising for future developments, offering potential to create dynamic narratives, influence non-player character (NPC) behaviour, and generate personalised environments that enhance player engagement.

Looking ahead, multifaceted approaches that integrate both game control and user assessment, especially those leveraging multimodal data, will be crucial. Such strategies are expected to improve behaviour analysis, personalise rehabilitation, enhance disease prediction, and ultimately increase patient engagement and adherence, thereby optimizing healthcare outcomes.

## CRediT authorship contribution statement

**Xiya Tao:** Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Nicolás Sáenz-Lechón:** Writing – review & editing. **Martina Eckert:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT to improve the language and writing style. After using this tool, the author (s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xiya Tao reports financial support was provided by Chinese Scholarship Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chbr.2025.100696.

## Data availability

No data was used for the research described in the article.

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